COLREGs-compliant model predictive collision avoidance for autonomous ships in restricted environments

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Abstract

Collision avoidance is integrated into a Model Predictive Control (MPC) framework by incorporating track constraints into the MPC's objective function. These constraints are governed by COLREGs to make rule-compliant decisions and actions for resolving track conflicts. The predictive model employed in MPC also allows for more accurate collision risk assessments, with safety distances calculated at each prediction step to avoid ship domain conflicts. Collision scenarios are classified into 16 categories (4 by 4 matrix), considering both the relative bearings of the own ship and the target ship, to ensure consistency in the decisions made by the involved ships. A series of experimental model tests were conducted at the Towing Tank for Manoeuvres in Confined Water, Antwerp, Belgium to validate the proposed controller. Initially, scenarios involving collision avoidance with static obstacles in combination with various ship speeds were examined, with varying safety margin parameters to impose different safety requirements. The obstacles were set both physical and virtual to investigate the influence of ship-obstacle interactions. Further tests involved a dynamic obstacle, creating overtaking and meeting scenarios with different speed combinations for the own ship and the dynamic obstacle. Simulations were performed to evaluate scenarios that are

impractical to replicate in the towing tank, thereby complementing experimental testing. The test results confirmed that collision avoidance can be successfully accomplished within a confined environment.

Keywords: COLREGs; Collision avoidance; Model predictive control; Free running test; Shallow water

1 Introduction

An autonomous ship's reference voyage can be scheduled by global path planning (Cao et al., 2022; P. Chen et al., 2020), which generates a collision-free route considering factors such as safety, cost and comfort. He et al. (2024) proposed a global path planning method tailored for inland waterways based on Fast Marching (FM) algorithm. In the method, the safety was satisfied by considering the distance to waterway borders and under keel clearance; a shortest route was then found after safety considerations, and the comfort of sailing was improved by modifying the optimal route with a so-called Dubins Model Filter to reduce track oscillations and fit manoeuvrability. However, a global route is not adaptive to dynamic obstacles or other new emerging obstacles, therefore a ship's track must be adjusted locally to avoid collision (Chiang and Tapia, 2018; Zhang et al., 2023) when a high collision risk is detected. Challenges are posed in automated collision avoidance at several aspects: increased traffic density in inland waterways, timing and extent of collision avoidance actions, and compliance to navigation rules.

Firstly, ships navigating on inland waterways encounter more frequent interactions with other vessels in proximity. Their manoeuvres are restricted by banks on either side, necessitating minimal deviation from the original reference route to avoid collisions with either the bank or the encountering vessel. Existing research typically uses conservative ship contours (e.g. circles, ellipses) and safety distances to obstacles for simplicity and sufficient safety margin. However, this might not be full applicable in restricted waterways due to limited navigable space. A ship's contour, along with surrounding safety margins, forms a ship domain. The ship domain concept appears frequently in collision avoidance research and is defined as a two-dimensional area surrounding a ship which other ships must avoid' (Fujii and Tanaka, 1971). A large variety of ship domain shapes are documented in the literature, and a few examples are shown in

Figure 1. The focus of collision avoidance can then be rephrased as preventing overlaps between ship domains. The ship domain adopted in this research is more similar to Rawson's model (Rawson et al., 2014) as presented in Figure 1, where the ship contour is more realistic, future ship contours are considered by extending the contour forward, and safety margins around the contour are included. In this research, rectangles will be utilized to represent ship contours, with additional safety margins defined in the longitudinal and the lateral directions. The ship domain extends forward along the predicted trajectory, which can more accurately reflect the necessary distance to remain collision-free, thereby saving space for collision avoidance manoeuvres and reducing additional control efforts.

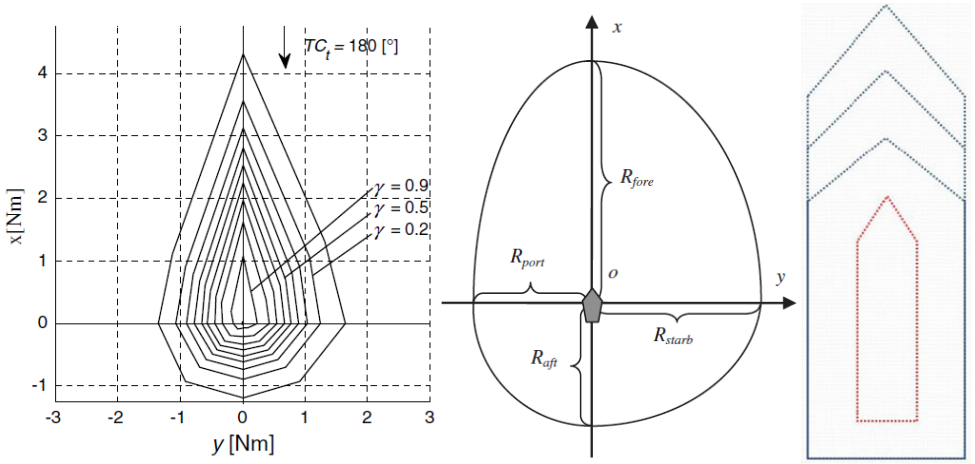


Figure 1 Ship domains proposed in literature. Left to right: Fuzzy domain (Pietrzykowski and Uriasz, 2009), quaternion domain (Wang, 2010), domain with nose extending forward depending on type of vessel and speed over ground (Rawson et al., 2014).

Additionally, the timing and extent of collision avoidance actions are closely related to collision risk assessment. A widely-used risk evaluation criterion is the closest point of approach (CPA), which refers to the estimated position where the own ship gets closest to the encountering moving vessel. Two related indices, *TCPA* and *DCPA*, were proposed by Kearon (1977) and are commonly used for risk assessment. TCPA is the consumed time to reach CPA from the present position, while *DCPA* is the distance between the two ships when they reach their respective CPAs. Based on assessed collision risk, decisions can be made on whether additional actions should be taken to keep a collision-free route and the extent of these actions. For instance, Brcko et al. (2021) employed a Fuzzy logic method to process CPA indices and

provided course alterations for collision avoidance. However, optimizing the mapping between control actions and risk assessment indices remains challenging to balance track deviation and safety.

Furthermore, ship navigation must adhere to navigation rules and conventions to avoid traffic chaos and reduce accidents. The maritime transportation is primarily governed by the International Regulations for Preventing Collisions at Sea (COLREGs) (IMO, 1972), which serves as the global standard to ensure safe navigation. These rules apply to all ships upon the seas or in all waters connected to them that are accessible to seagoing ships. In contrast, inland navigation is subject to regional and local laws, which vary between countries and waterway authorities. However, they are generally modelled after COLREGs and share many of the same basic principles. The main distinctions in inland navigation laws typically result from unique environmental circumstances, such as the confined spaces of rivers and canals and larger traffic density, which may call for additional regulations to prevent collisions. To develop a flexible collision avoidance mechanism that can be applied to a wide range of scenarios, this study mainly follows the provisions set out by COLREGs, which still allows for adjustments to meet regional regulatory requirements when necessary. COLREGs define three typical collision scenarios: overtaking, head-on and crossing. However, identifying these scenarios can be ambiguous with statements in the rules, which complicates the development of corresponding algorithms where exact indices and bounds are required. In literature, various means can be found. For instance, in (Lyu and Yin, 2019; Tam and Bucknall, 2013), the relative bearing was adopted to divide the region around the own ship into four sectors, and each sector represents a collision scenario. Du et al. (2021) divided the ranges of the relative bearing and relative heading respectively into 8 groups, and consequently 64 combinations are obtained and identified as various collision scenarios. However, these methods have some issues and limitations, as analysed by García Maza and Argüelles (2022). Correctly identifying collision scenarios is crucial since different decisions and actions are defined for each scenario. Misunderstanding happens with wrongly identified scenarios, hence increasing danger. This research will consider the consistency between the decisions of only two involved ships. The relative bearings with respect to the own ship and the target ship (encountering ship) are both utilized as criteria. In the process of

designing scenario identification approaches, ensuring consistent identification results when switching roles of the two ships, without conflicts, guarantees a unique scenario code and eliminates misunderstandings. For instance, if the own ship is identified as overtaking the target ship, the target ship must be identified as being overtaken when applying the same approach. If a head-on or crossing situation is occurring for the own ship, the same situation must also happen to the target ship. More details will be provided in Section 3.

Resolving local track conflicts can be achieved by replanning the reference route (Lazarowska, 2015; Liu et al., 2017) or taking additional control actions to resolve track conflicts (Abdelaal et al., 2018; Brcko et al., 2021). Applying path planning algorithms to reschedule routes based on detected situations proposes a very high requirement in terms of calculation efficiency. Langxiong et al. (2025) designed a real-time path planning algorithm based on an improved artificial potential field method, considering the collision risk for ship encounters. It was proved that the technique can be applied in restricted waterways and in both static and dynamic scenarios to solve track conflicts with compliance to navigation rules. However, collision avoidance can be more elegantly realized by adopting another strategy: taking additional control actions to deviate from the original route to avoid collisions according to navigation regulations, and resuming the initial voyage after eliminating collision risks. Especially, it can be seamlessly integrated into a Model Predictive Control (MPC) framework, optimizing decisions and actions for resolving conflicts. MPC's capability of incorporating constraints in its multiple-objective optimization offers opportunities to consider track constraints regarding encountering obstacles with high risk of collision. Additionally, MPC employs a predictive model to forecast the own ship's manoeuvring behaviours. This model can also be used for evaluating collision risks to encountering obstacles based on the predicted own ship's future trajectory, potentially improving the accuracy of risk evaluation. Most existing risk evaluation models are overly simplistic that only consider ship kinematics, but ship dynamics significantly influences the ship's track and brings notable differences in risk evaluation. Recent studies on MPC-based collision avoidance, such as those by Hagen et al. (2018) and Eriksen and Breivik (2017), have made contributions to the advancement of maritime collision prevention technologies. A comprehensive review of MPC applied to ship track control

can be found in a previous study by the authors (He et al., 2023). Building upon this foundation, the present research applies MPC to achieve COLREGs-compliant collision avoidance. In doing so, it introduces novel models for defining safety distance definition, collision scenario identification, risk evaluation and decision making. Different objective functions, safety requirements, predictive models and control strategies are tested with extensive experimental tests, proving the effectiveness of proposed design.

2 Model predictive control

2.1 Problem formulation

In this study, automated collision avoidance is achieved by incorporating collision constraints into the MPC's optimization problem (He et al., 2023):

$$\begin{aligned} &\operatorname{argmin} J(\boldsymbol{x},\boldsymbol{u}) = \sum_{i=1}^{N_\mathrm{p}} \left(\boldsymbol{x}_i - \boldsymbol{x}_i^{\mathrm{ref}}\right)^{\mathrm{T}} \boldsymbol{Q} \left(\boldsymbol{x}_i - \boldsymbol{x}_i^{\mathrm{ref}}\right) \\ &\text{subject to:} \\ &\mathrm{C1}: \boldsymbol{x}_{i+1} = f(\boldsymbol{x}_i, \boldsymbol{u}_i) \\ &\mathrm{C2}: \boldsymbol{u}_{\min} \le \boldsymbol{u}_i \le \boldsymbol{u}_{\max} \\ &\mathrm{C3}: \Delta\boldsymbol{u}_{\min} \le \Delta\boldsymbol{u}_i \le \Delta\boldsymbol{u}_{\max} \\ &\mathrm{C4}: \left(x_i - x_{j,i}^{\mathrm{obs}}\right)^2 + \left(y_i - y_{j,i}^{\mathrm{obs}}\right)^2 \ge D_{j,i};\ i = 1, \dots, N_\mathrm{p}; j = 1, \dots, N_{\mathrm{obs}} \end{aligned} \tag{1}$$

where $J(\boldsymbol{x},\boldsymbol{u})$ is the objective function, $\boldsymbol{Q}$ is the weight matrix, $f(\boldsymbol{x},\boldsymbol{u})$ represents the ship dynamics (discretely calculated by using Euler method in this research), and at time step i, $\boldsymbol{x}_i$ and $\boldsymbol{x}_i^{\mathrm{ref}}$ are the predicted and reference ship state vectors, $\boldsymbol{u}_i$ and $\Delta\boldsymbol{u}_i$ are the input vector and its increment, x_i and y_i represent the predicted x- and y-positions of the ship, $x_{j,i}^{\mathrm{obs}}$ and $y_{j,i}^{\mathrm{obs}}$ are the predicted positions of the obstacle j, $D_{j,i}$ is the safe distance between the own ship and the obstacle j, N_p is the prediction horizon, and N_{obs} is the number of obstacles. The ship state vectors and the input vector are composed of as follows:

$$\boldsymbol{x}_i = [x_i, y_i, \chi_i, u_i, v_i, r_i]^{\mathrm{T}}$$

$$\boldsymbol{x}_i^{\mathrm{ref}} = \left[p_x(\varpi_i), p_y(\varpi_i), p_\chi(\varpi_i), p_u(\varpi_i), p_v(\varpi_i), p_r(\varpi_i)\right]^{\mathrm{T}} \tag{2}$$

$$\boldsymbol{u}_i = [\delta_i, n_i]^{\mathrm{T}}$$

where χ_i, u_i, v_i and r_i denote the predicted ship course, surge velocity, sway velocity and yaw rate respectively, and δ_i and n_i are the rudder angle and propeller rate. More details about the MPC, including the reference generation and optimization solver, can be consulted in the authors' previous research (He et al., 2023). The objective function is subject to four types of constraints, C1 to C4. C1 represents the ship dynamics constraint, and C2 and C3 are constraints of actuators' physical capabilities. C4 describes the constraints of collision avoidance, which will be elaborated in Section 3. Without considering C4, the controller returns to a path following or trajectory tracking controller.

2.2 Predictive model

The 3-DOF (Degrees of Freedom) ship manoeuvring motion in the horizontal plane can be described as:

$$\begin{aligned} \boldsymbol{x} &= [\boldsymbol{\eta}; \boldsymbol{v}] + [0; 0; \beta; 0; 0; 0] \\ \dot{\boldsymbol{\eta}} &= \boldsymbol{R}(\psi)\boldsymbol{v} \\ \dot{\boldsymbol{v}} &= \boldsymbol{f}(\boldsymbol{v}, \boldsymbol{u}) \\ \boldsymbol{u} &= [\delta; n] \end{aligned} \tag{3}$$

where $\boldsymbol{\eta}$ denotes the position and orientation vector, $\boldsymbol{v}$ denotes the velocity vector, β is the drift angle, and $\boldsymbol{R}(\psi)$ is the rotation matrix in the horizontal plane. Two kinds of predictive models are designed (He et al., 2025b):

a) Speed adaptive Nomoto model

$$\dot{\boldsymbol{v}} = \begin{bmatrix} 0 \\ 0 \\ -\dfrac{r}{T} + \dfrac{K}{T}\delta - \dfrac{B}{T} \end{bmatrix} \tag{4}$$

$$T = c_1 U^{-1}, B = c_2 U + c_3, K = c_4 U$$

where K, T and B (rad/s) are the parameters of the nonlinear Nomoto model, and c_1 to c_4 are the coefficients to adjust the model parameters according to the ship speed U.

b) 3-DOF simplified model

$$\dot{\boldsymbol{v}} = \begin{bmatrix} a_1 u + a_2 vr + a_3 n \\ b_1 ur + b_2 v + b_3 r \\ c_1 Ur + c_2 U^2 + c_3 U + c_4 Un\delta \end{bmatrix} \tag{5}$$

where a_1 to a_3, b_1 to b_3, and c_1 to c_4 are the model coefficients. The model coefficients are estimated by a linear least squares method, driven by manoeuvring data collected in free running tests (acceleration and zigzag tests).

2.3 Speed adaptive control

The prediction horizon is adapted with the ship speed, propeller rate or both, as listed in Table 1, where four speed adaptive schemes are proposed. For the collision avoidance scenario, in order to let the controlled ship react early, the prediction horizon is increased compared to the basic used in path following or trajectory tracking. This is achieved by the following equation:

$$N_{\mathrm{p}} = \begin{cases} K_{\mathrm{OB}} N_{\mathrm{p0}}, \text{if performing collision avoidance} \\ K_{\mathrm{NP}} N_{\mathrm{P0}}, \text{else if } |T_{\mathrm{e}}| > K_{\mathrm{te}} B \text{ or } |H_e| > K_{\mathrm{he}} \\ N_{\mathrm{P0}}, \quad \text{else} \end{cases} \tag{6}$$

where N_{P0} is the basic prediction horizon and K_{OB} and K_{NP} are tuneable positive gains that should not be less than 1, T_{e} and H_e are the cross-track error and heading error respectively, K_{te} and K_{he} are nondimensional tolerances of the cross-track error and the heading error respectively, and B is the ship beam (m).

Table 1. Speed adaptive scheme for prediction horizon. λ is a proportional factor that is suggested to be set within [0.75, 1]. Recommendation for setting λ_1 and λ_2: $\lambda_1 = 0.75$, $\lambda_2 = (1 - 0.75)/ (U_{\max} - U_{\min})$, where $U_{\max}$ and $U_{\min}$ are respectively the upper and the lower bounds of the ship speed.

Speed adaptive scheme	Formula

1	$N_{\mathrm{p0}} = \lambda \frac{L}{U}$
2	$N_{\mathrm{p0}} = \lambda \frac{L}{K_u n}$
3	$N_{\mathrm{p0}} = \lambda \frac{L}{\sqrt{K_u U n}}$
4	$N_{\mathrm{p0}} = \lambda \frac{L}{\sqrt{K_u U n}}$
	$\lambda = \lambda_1 + \lambda_2 (U - U_{\min})$

3 Compliance with COLREGs

3.1 Overview of COLREGs

COLREGs define three typical vessel interaction scenarios: overtaking (Rule 13), head-on (Rule 14) and crossing (Rule 15). Rules 16 and 17 specify the responsibilities of give-way and stand-on vessel. These rules are illustrated in Figure 2, and are elaborated below:

(1) Overtaking: Rule 13 states that a vessel is overtaking when approaching another from more than 22.5 degrees abaft her beam. The overtaking vessel must keep clear, while the vessel being overtaken should maintain course and speed unless collision risk arises (Rule 17). Although Rule 13 doesn't specify passing side, inland waterways — where COLREGs are still applicable if connected to seas — the port-side is typically preferred for overtaking. For instance, in Belgium, Article 13 in the navigation rules for the Lower Scheldt (Scheepvaartreglement voor de Beneden-Zeeschelde) states that overtaking vessels must pass on the port side of the vessel being overtaken if conditions allow. Therefore, a preference is assigned to the port side for overtaking manoeuvres.

(2) Head-on: Rule 14 covers situations where two power-driven vessels meet on reciprocal or nearly reciprocal courses with collision risk. Both vessels must alter course to starboard, passing on each other's port side.

(3) Crossing: Rule 15 applies to crossing situations that are neither overtaking nor head-on and involve risk of collision. The vessel with another on her starboard side must give way, usually by altering

course to starboard to pass behind. Crossing scenarios are divided into two subgroups: crossing A (own ship gives way) and crossing B (own ship stands on), as illustrated in Figure 2.

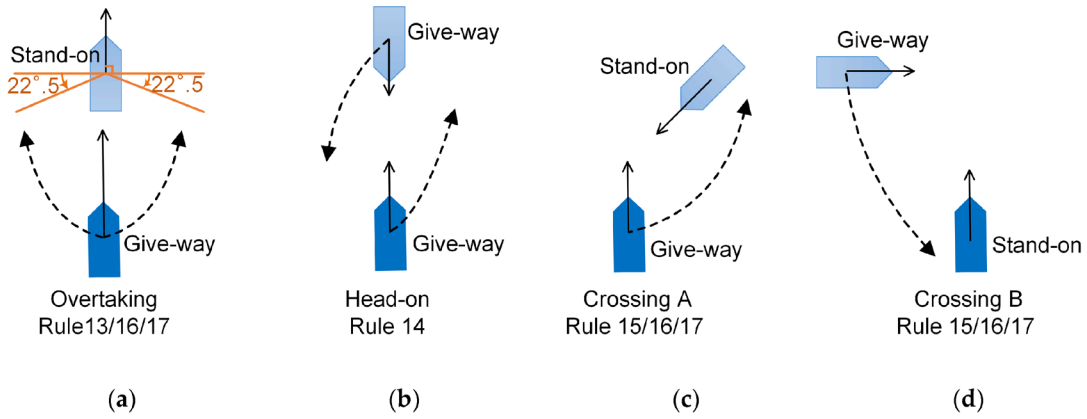


Figure 2. Typical collision scenarios and suggested manoeuvres from the COLREGs (Lyu and Yin, 2018)

3.2 Risk evaluation

Rule 7 of COLREGs states that each vessel shall use appropriate means to determine whether collision risk exists. The following shows how to evaluate the risk based on trajectory prediction of the own ship and the other ship. In order to reduce calculation costs, only the other ships within an alert circle with a radius of R_a are taken into consideration. For the obstacles within the alert area, collision risk is evaluated to decide whether action should be carried out to avoid collision. The MPC employs a predictive model to forecast the future trajectory within a prediction horizon, and it can also be adopted here to generate trajectory prediction for the own ship by applying the current rudder angle and propeller rate as inputs to the predictive model.

The own ship only needs to take collision avoidance action when she becomes the give-way ship as defined in the COLREGs. Therefore, it is of more interest to assume that the other ship is the stand-on ship. Rule 17 states the action by the stand-on vessel: 'where one of two vessels is to keep out of the way the other shall keep her course and speed'. The other ship's trajectory can be easily predicted once knowing her current position, course and speed:

$$x_{j,i}^{\text{obs}} = x_{j,i-1}^{\text{obs}} U_j^{\text{obs}} \cos \chi_j^{\text{obs}}$$
$$y_{j,i}^{\text{obs}} = y_{j,i-1}^{\text{obs}} U_j^{\text{obs}} \sin \chi_j^{\text{obs}} \tag{7}$$

If existing a prediction step i where the following condition is satisfied:

$$\left(x_i - x_{j,i}^{\text{obs}}\right)^2 + \left(y_i - y_{j,i}^{\text{obs}}\right)^2 < D_{j,i} \tag{8}$$

the risk of collision is evaluated as high. Otherwise, the risk of collision is low. To ensure the other vessel is finally past and clear, an additional constraint is supplemented when reducing the risk level from high to low:

$$\left(x_i - x_{j,i}^{obs}\right)^2 + \left(y_i - y_{j,i}^{obs}\right)^2 > K_D D_{j,i} \tag{9}$$

where $K_D > 1$. Figure 3 illustrates the evaluated risks for two scenarios. In the left sub-figure, the predicted trajectories of the own ship and the other ship can keep safety distances everywhere, which involves a low risk of collision. In the right sub-figure, the two trajectories have an intersection (at several points, the safety distances are violated), and therefore a high risk of collision is involved.

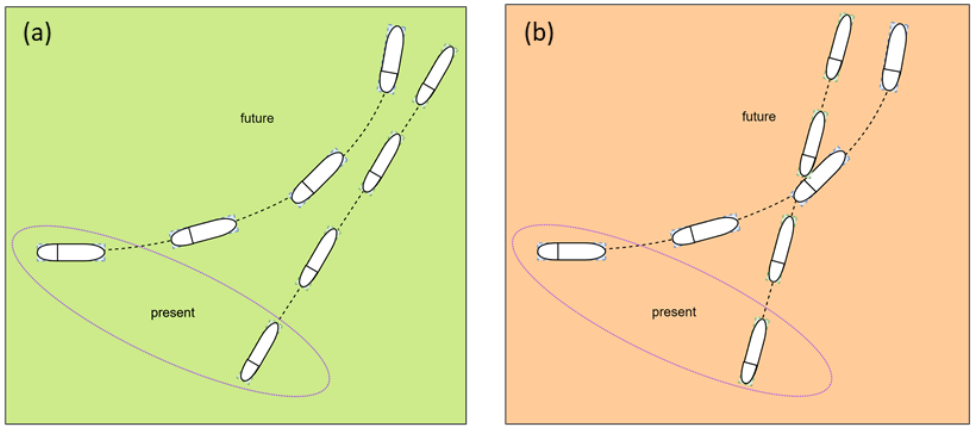


Figure 3. Risk evaluation (a) low risk of collision, (b) high risk of collision. The other ship keeps her speed and course.

3.3 Safety distance

Rule 8 of COLREGs states that action performed to avoid collision shall keep a safety distance with the other vessel. In inland waters, as the navigable space is very limited and ships usually sail close to each

other, the safety distance cannot be too conservative. Otherwise, the collision avoidance manoeuvres cannot be achieved within the navigable area.

To avoid conflict of the two ships, their geometric areas should not overlap. Furthermore, to improve the safety, for instance to reduce the impact of ship-ship interaction forces that may attract both ships towards each other, an additional margin may be necessary. The geometric areas together with additional margins form the ship domain, of which various shapes can exist as introduced in Section 1. For simplification, the geometric area of a ship is regarded as a rectangle and it is assumed that the centre of gravity is located in the geometric centre (this assumption needs to be made as in most cases the position of the centre of gravity of the other ship is unknown to the own ship), as shown in Figure 4.

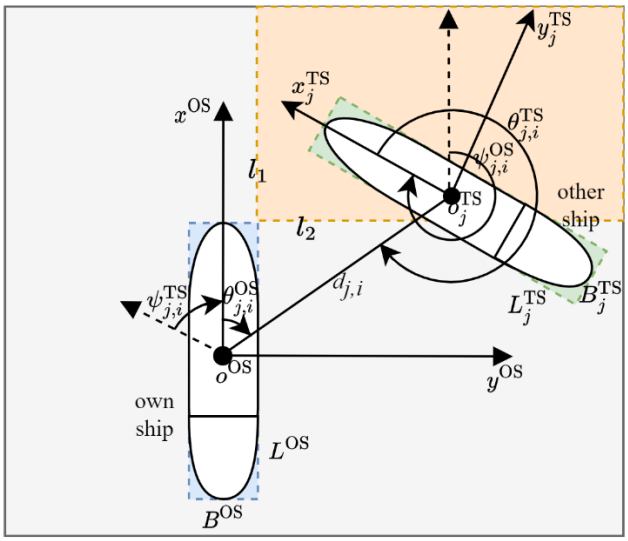


Figure 4. Calculating the safety distance for the j-th obstacle (target ship) at the i-th prediction step.

In this figure, two coordinate systems are involved: $o^{\mathrm{OS}}x^{\mathrm{OS}}y^{\mathrm{OS}}$ is the own ship body-fixed frame, and $o_j^{\mathrm{TS}}x_j^{\mathrm{TS}}y_j^{\mathrm{TS}}$ is the target ship body-fixed frame. $\psi_{j,i}^{\mathrm{OS}}$ and $\theta_{j,i}^{\mathrm{OS}}$ are respectively the relative heading and the relative bearing of the own ship (expressed in $o^{\mathrm{OS}}x^{\mathrm{OS}}y^{\mathrm{OS}}$), while $\psi_{j,i}^{\mathrm{TS}}$ and $\theta_{j,i}^{\mathrm{TS}}$ are respectively the relative heading and the relative bearing of the target ship (expressed in $o_j^{\mathrm{TS}}x_j^{\mathrm{TS}}y_j^{\mathrm{TS}}$). L^{OS} and B^{OS} are the length and the breadth of the own ship, L_j^{TS} and B_j^{TS} are the length and breadth of the j-th target ship, $d_{j,i}$ is the distance between the ship and the target ship. To keep a safe distance between the two ship domains, there

should be enough lateral margin or enough longitudinal margin. At least one of the following conditions, Eq. (10) and Eq. (11), should be satisfied:

$$\begin{aligned} d_{j,i}\left|\sin\theta_{j,i}^{\mathrm{OS}}\right| &> k_{\mathrm{B}}d_{\mathrm{Bc}} + d_{\mathrm{Ba}} \\ d_{\mathrm{Bc}} &= \frac{L_j^{\mathrm{TS}}\left|\sin\psi_{j,i}^{\mathrm{OS}}\right| + B_j^{\mathrm{TS}}\left|\cos\psi_{j,i}^{\mathrm{OS}}\right| + B^{\mathrm{OS}}}{2} \end{aligned} \tag{10}$$

$$\begin{aligned} d_{j,i}\left|\cos\theta_{j,i}^{\mathrm{OS}}\right| &> k_{\mathrm{L}}d_{\mathrm{Lc}} + d_{\mathrm{La}} \\ d_{\mathrm{Lc}} &= \frac{L_j^{\mathrm{TS}}\left|\cos\psi_{j,i}^{\mathrm{OS}}\right| + B_j^{\mathrm{TS}}\left|\sin\psi_{j,i}^{\mathrm{OS}}\right| + L^{\mathrm{OS}}}{2} \end{aligned} \tag{11}$$

where d_{Bc} and d_{Lc} are the minimum lateral distance and the minimum longitudinal distance respectively to solve any conflict of the own ship domain and the other ship domain, k_{B} and k_{L} are two correction factors that are larger than 1 to allow a tolerance for the mismatch of the centre of gravity and the geometric centre, and d_{Ba} and d_{La} are the additional lateral and longitudinal distances to improve safety. k_{B} and k_{L} can also be used to control the margin. To satisfy Eq. (10) or Eq. (11) or both, the distance between the two ships should be:

$$\begin{aligned} d_{j,i} &> D_{j,i} \\ D_{j,i} &= \min\left\{\frac{k_{\mathrm{B}}d_{\mathrm{Bc}} + d_{\mathrm{Ba}}}{\left|\sin\theta_{j,i}^{\mathrm{OS}}\right|}, \frac{k_{\mathrm{L}}d_{\mathrm{Lc}} + d_{\mathrm{La}}}{\left|\cos\theta_{j,i}^{\mathrm{OS}}\right|}\right\} \end{aligned} \tag{12}$$

where $D_{j,i}$ denotes the safety distance. However, when the own ship domain crosses both the line l_1 and the line l_2 (as shown in Figure 4), the distance will always be insufficient according to Eq. (12), even if there is no overlap of the two ship domains (the additional margin is not considered here). In this case, if the roles of the own ship and the target ship are switched, Eq. (12) will be satisfied, and the safety distance becomes

$$\begin{aligned} D_{j,i} &= \min\left\{\frac{k_{\mathrm{B}}\bar{d}_{\mathrm{Bc}} + d_{\mathrm{Ba}}}{\left|\sin\theta_{j,i}^{\mathrm{TS}}\right|}, \frac{k_{\mathrm{L}}\bar{d}_{\mathrm{Lc}} + d_{\mathrm{La}}}{\left|\cos\theta_{j,i}^{\mathrm{TS}}\right|}\right\} \\ \bar{d}_{\mathrm{Bc}} &= \frac{L^{\mathrm{OS}}\left|\sin\psi_{j,i}^{\mathrm{TS}}\right| + B^{\mathrm{OS}}\left|\cos\psi_{j,i}^{\mathrm{TS}}\right| + B_j^{\mathrm{TS}}}{2} \end{aligned} \tag{13}$$

$$\bar{d}_{\mathrm{Lc}} = \frac{L^{\mathrm{OS}} \left|\cos \psi_{j,i}^{\mathrm{TS}}\right| + B^{\mathrm{OS}} \left|\sin \psi_{j,i}^{\mathrm{TS}}\right| + L_j^{\mathrm{TS}}}{2}$$

Therefore, to unify the solution, the safety distance is modified to the smaller of the results calculated by Eq. (12) and Eq. (13) as:

$$D_{j,i} = \min\left\{\frac{k_{\mathrm{B}} d_{\mathrm{Bc}} + d_{\mathrm{Ba}}}{\left|\sin \theta_{j,i}^{\mathrm{OS}}\right|}, \frac{k_{\mathrm{L}} d_{\mathrm{Lc}} + d_{\mathrm{La}}}{\left|\cos \theta_{j,i}^{\mathrm{OS}}\right|}, \frac{k_{\mathrm{B}} \bar{d}_{\mathrm{Bc}} + d_{\mathrm{Ba}}}{\left|\sin \theta_{j,i}^{\mathrm{TS}}\right|}, \frac{k_{\mathrm{L}} \bar{d}_{\mathrm{Lc}} + d_{\mathrm{La}}}{\left|\cos \theta_{j,i}^{\mathrm{TS}}\right|}\right\} \tag{14}$$

This provides a more accurate solution and also unifies the solutions for the involved ships.

3.4 Scenario identification

As aforementioned, three typical collision scenarios are included in COLREGs, and in each scenario, different actions need to be performed by the involved ships. Therefore, to reach a decision for collision avoidance, the present collision scenario should be first identified. In Rule 13, it is clearly stated that the relative bearing ('another vessel from a direction more than 22.5 degrees abaft her beam') should be utilized for identifying the overtaking scenario. From this point of view, the relative bearing acts as the index for scenario identification in this research. The relative bearing can be both expressed in the own ship body-fixed coordinate system (θ_j^{OS}) and in the other ship body-fixed coordinate system (θ_j^{TS}), as shown in Figure 5. θ_j^{TS} needs to be used for identifying whether the own ship is overtaking the other ship, while θ_j^{OS} needs to be employed for checking if the own ship is being overtaken by the other ship. Consequently, both indexes are significant in scenario identification and will be applied in this research. They are calculated as:

$$\begin{aligned} \theta_j^{\mathrm{OS}} &= \mathrm{atan2}\left(y_j^{\mathrm{obs}} - y, x_j^{\mathrm{obs}} - x\right) - \psi \\ \theta_j^{\mathrm{TS}} &= \mathrm{atan2}\left(y - y_j^{\mathrm{obs}}, x - x_j^{\mathrm{obs}}\right) - \psi_j^{\mathrm{obs}} \end{aligned} \tag{15}$$

Usually, the heading and the course are very close to each other. However, in order to consider the special scenario that the ship is reversing, the course can be adopted in Eq.(15), and it gives

$$\begin{aligned} \theta_j^{\mathrm{OS}} &= \mathrm{atan2}\left(y_j^{\mathrm{obs}} - y, x_j^{\mathrm{obs}} - x\right) - \chi \\ \theta_j^{\mathrm{TS}} &= \mathrm{atan2}\left(y - y_j^{\mathrm{obs}}, x - x_j^{\mathrm{obs}}\right) - \chi_j^{\mathrm{obs}} \end{aligned} \tag{16}$$

The scenario identification will only be performed when a high risk of collision is detected (see Section 3.2). Figure 5 illustrates the sector areas, OS-A, OS-B, OS-C, and OS-D, around the own ship that are divided by utilizing $\theta_{j,i}^{\mathrm{OS}}$, which assists scenario identification. The similar sector areas, TS-A, TS-B, TS-C, and TS-D, are also defined for the other ship. In this figure, θ_0 is an angle that is adopted for identifying the head-on scenario. Since Rule 14 does not provide a number, this parameter should be set by negotiating with different parties. Rule 14 only describes the head-on scenario by using 'reciprocal or nearly reciprocal courses'. To create this scenario, $\theta_{j,i}^{\mathrm{OS}}$ and $\theta_{j,i}^{\mathrm{TS}}$ should be both restricted to the vicinity of zero, and here it is defined to $[-\theta_0, \theta_0]$, where θ_0 should be a sufficiently small angle.

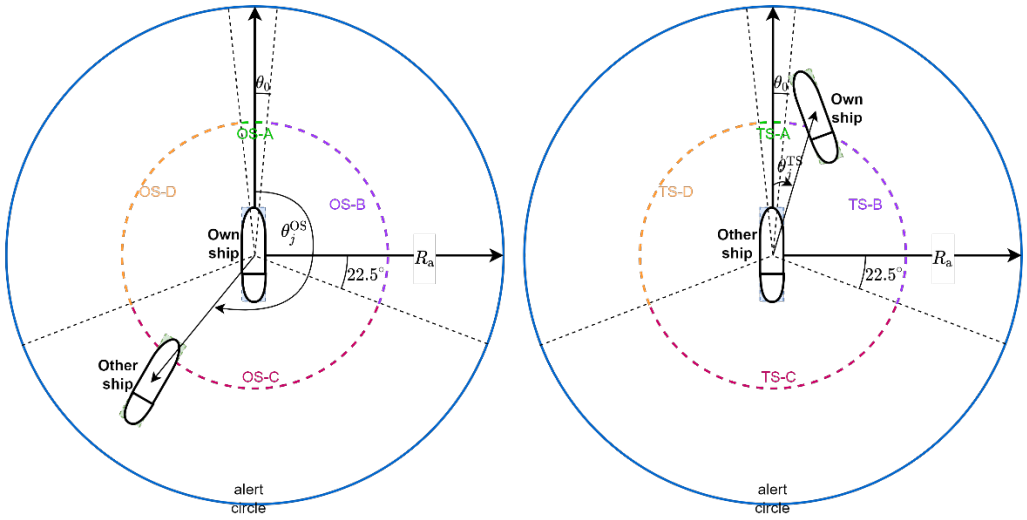


Figure 5. Sector areas around the own ship divided on the basis of the relative bearing expressed in the own ship body-fixed coordinate system for identifying collision scenarios.

Table 2. Scenario identification and corresponding actions required for involved ships. TS and OS respectively denote the other ship and the own ship. SB, PS and SO respectively denotes the give-way directions, starboard side, port side and stand-on.

	OS sector areas	OS-A	OS-B	OS-C	OS-D
TS sector areas	θ_j^{OS} / θ_j^{TS}	$[-\theta_0, \theta_0]$	$(\theta_0, 112.5°]$	$(112.5°, 180°]$ $\cup$ $[-180°, -112.5°)$	$[-112.5°, -\theta_0)$
TS-A	$[-\theta_0, \theta_0]$	head-on OS = SB TS = SB	crossing A OS = SB TS = SO	overtaken OS = SO TS = PS	crossing B OS = SO TS = SB
TS-B	$(\theta_0, 112.5°]$	crossing B OS = SO TS = SB	no risk	overtaken OS = SO TS = PS	crossing B OS = SO TS = SB

TS-C	$(112.5°, 180°]$ $\cup [-180°, -112.5°)$	overtaking OS = PS TS = SO	overtaking OS = PS TS = SO	no risk	overtaking OS = PS TS = SO
TS-D	$[-112.5°, -\theta_0)$	crossing A OS = SB OS = SO	crossing A OS = SB OS = SO	overtaken OS = SO TS = PS	no risk

Table 2 lists all possible combinations of $\theta_{j,i}^{\mathrm{OS}}$ and $\theta_{j,i}^{\mathrm{TS}}$ and their pointing scenarios as well as the give-way directions for the own ship and the other ship. Figure 6 provides an illustration for the scenarios generated in Table 2. It is very clear that head-on only happens in the scenario S01, overtaking happens when the own ship in the sector area TS-C (scenarios S09, S10, S11 and S12), and being overtaken happens when the target ship is in the sector area OS-C (scenarios S03, S07, S11 and S15). The overtaking scenarios and the being overtaken scenarios overlap at S11. However, this scenario is usually created when the two ships have finished the meeting, and it will not be identified since no risk of collision is involved, as shown in Figure 6. The remaining is crossing scenarios. S06 and S16 will not be identified as there is also no risk of collision involved. It is also interesting to check the duality in the table in Figure 6. The table is used for the own ship to identify the collision scenario, and the transposed table can be adopted to identify the collision scenario for the other ship. When a ship is meeting another ship, the other ship is also meeting this ship. When a ship is overtaking another ship, another ship is being overtaken. When a ship is crossing another ship, another ship is also crossing this ship. Therefore, the following condition should be kept:

$$\begin{gathered} \boldsymbol{S}(m,n) = \text{head-on} \rightarrow \boldsymbol{S}(n,m) = \text{head-on}; \\ \boldsymbol{S}(m,n) = \text{overtaking} \rightarrow \boldsymbol{S}(n,m) = \text{being overtaken}; \\ \boldsymbol{S}(m,n) = \text{crossing (A or B)} \rightarrow \boldsymbol{S}(n,m) = \text{crossing (A or B)} \end{gathered} \tag{17}$$

where $\boldsymbol{S}$ denotes the collision scenario matrix (table), m and n are respectively the row index and the column index. Furthermore, in a collision scenario, the own ship and the other ship are expected to take specific actions to avoid collision, and if the role of the two ships is switched, the required action should be switched accordingly:

$$\boldsymbol{A}_{\mathrm{OS}}(m,n) = \boldsymbol{A}_{\mathrm{TS}}(n,m) \tag{18}$$

$$\boldsymbol{A}_{\mathrm{TS}}(m,n) = \boldsymbol{A}_{\mathrm{OS}}(n,m)$$

where $\boldsymbol{A}_{\mathrm{OS}}$ and $\boldsymbol{A}_{\mathrm{TS}}$ respectively denote the action required for the own ship and the action required for the other ship. It is seen that Eq. (17) and Eq. (18) are satisfied in the table in Figure 6, implying that misunderstanding between the two ships can be avoided by applying for identifying collision scenarios and assigning corresponding required actions for collision avoidance. It should be mentioned that the background color in Figure 6 reflects the frequency level that the scenario may happen according to the authors' knowledge and the mathematic view (the ease to create the scenario with a collision risk); the background colour is only used for illustration.

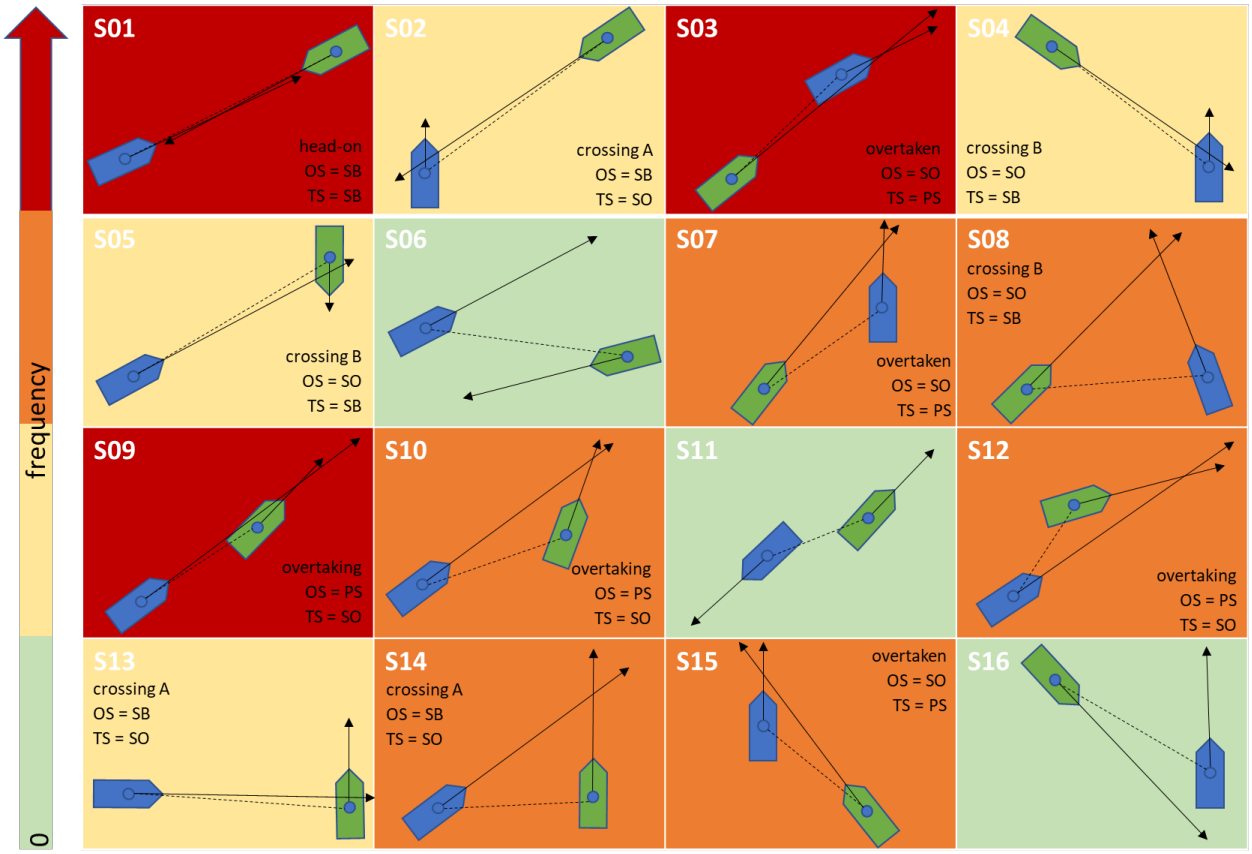


Figure 6. Collision scenarios identified according to the relative bearings. Blue ship represents the own ship and green ship represents the other ship.

Here any moored ships will also be taken into account, the collision scenario with them is categorized into a new type, static obstacle, where the own ship is required to give way at any direction since this process is not regulated by COLREGs. To summarize, the following scenarios will be identified:

1) static obstacle: if $|U_{\mathrm{OS}}| = 0$

2) others: if $|U_{\mathrm{OS}}| > 0$

   a) overtaking: if $\theta_j^{\mathrm{TS}} \in (112.5°, 180°] \cup [-180°, -112.5°)$

   b) being overtaken: if $\theta_j^{\mathrm{OS}} \in (112.5°, 180°] \cup [-180°, -112.5°)$

   c) head-on: if $\theta_j^{\mathrm{TS}} \in [-\theta_0, \theta_0]$ and $\theta_j^{\mathrm{OS}} \in [-\theta_0, \theta_0]$

   d) crossing B: if $\theta_j^{\mathrm{OS}} \in [-112.5°, -10°)$ and $\theta_j^{\mathrm{TS}} \in [-\theta_0, 112.5°]$, or if $\theta_j^{\mathrm{OS}} = [-\theta_0, \theta_0]$ and $\theta_j^{\mathrm{TS}} \in (\theta_0, 112.5°]$

   e) crossing A: if $\theta_j^{\mathrm{TS}} \in [-112.5°, -10°)$ and $\theta_j^{\mathrm{OS}} \in [-\theta_0, 112.5°]$, or if $\theta_j^{\mathrm{TS}} = [-\theta_0, \theta_0]$ and $\theta_j^{\mathrm{OS}} \in (\theta_0, 112.5°]$

where U_{OS} is the speed of the own ship.

After a ship alters her course and speed to avoid collision, the collision scenario may be changed after a while. For instance, when a ship is overtaking another ship by using her port side and has passed the direction 22.5 degrees abaft her beam, the collision scenario becomes crossing A. Then the ship is required to turn to her starboard side to give way, which generates an unreasonable and stupid manoeuvre. To avoid this situation, Rule 13 states ‘any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these Rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear’. This means the collision scenario should not be altered once it has been identified until the collision risk is solved. Therefore, the strategy in this research becomes:

a) the risk of collision will be calculated at each time step.

b) if the risk is high and no scenario has been assigned, the collision scenario will be identified.

c) if the risk is low, the collision scenario will be reset to none.

3.5 Decision making for collision avoidance

Table 3 shows the framework of the MPC collision avoidance algorithm (MPC_CA). MPC_CA can solve collision avoidance with multiple encountering ships (obstacles). It loops the list that contains all obstacles. For each obstacle, MPC_CA first checks if the obstacle locates within the alert circle. For the inside obstacles, risk of collision with them will be evaluated. Collision scenarios with the obstacles with high risk are then identified and the corresponding give-way strategies are added into a list, *GWL*. After all obstacles have been looped, a final decision for giving way will be made by checking *GWL*. The priority is given to steering to starboard side since only in the overtaking scenario the preference is assigned to port side. However, this preference is based on the local rules in inland waterways and actually the COLREGs allows the overtaking ship to take any side. Therefore, it is reasonable to give priority to the starboard steering, and if *GWL* contains 'Give-way-SB', MPC_CA will set a constraint to steering to the starboard side. Otherwise, it will continue checking if *GWL* contains 'Give-way-PS', and it sets a constraint to steering to the port side if the condition is satisfied, and no constraint is set if not. An optimal solution is then provided to minimize the deviation from the planned path and in the meantime to satisfy the steering direction constraint.

Table 3 Pseudo-code of COLREGS-compliant MPC collision avoidance algorithm

Algorithm: COLREGs-compliant MPC collision avoidance

Input: ship states, configuration file (controller's parameters, waypoints)
Output: control actions
Apply speed adaptive strategy
$k = 1$; $GWL = \{\}$
while $k \leq N_{\text{ob}}$ **do**
  if $\sqrt{\left(x - x_k^{\text{obs}}\right)^2 + \left(y - y_k^{\text{obs}}\right)^2} < R_{\text{a}}$ **then**
    Evaluate collision risk for obstacle k
    if high collision risk **then**
      Identify collision scenario
      if obstacle is static **then**
        $GWL \leftarrow GWL \cup \{'\text{Give-way}'\}$
      else if head-on or crossing A situation **then**
        $GWL \leftarrow GWL \cup \{'\text{Give-way-SB}'\}$
      else if overtaking **then**
        $GWL \leftarrow GWL \cup \{'\text{Give-way-PS}'\}$
      else
        $GWL \leftarrow GWL \cup \{'\text{Stand-on}'\}$
      end if
    end if
  end if
  $k \leftarrow k + 1$
end while
if 'Give-way-SB' $\in GWL$ **then**
  if not already on starboard side **then**
    Set steering direction constraint to starboard side
  end if
else if 'Give-way-PS' $\in GWL$ **then**
  if not already on port side **then**
    Set steering direction constraint to port side
  end if
end if
Solve MPC optimization problem
Return optimal solution (control actions)

3.6 Modification of the MPC optimization problem

In the previous section, it is discussed that the final decision making in collision avoidance is about the constraint setting of the steering direction. As a constant rudder angle is maintained throughout the prediction horizon, the yaw rate is usually stabilized at the end of the prediction horizon, provided the prediction horizon is sufficiently long. A negative yaw rate implies the ship is steering to the port side, and vice versa. Therefore, the steering direction of the ship can be determined by examining the stabilized yaw

rate, and constraining the steering direction can be effectively managed by imposing limits on the stabilized yaw rate.

3.6.1 Constrained optimization

The MPC optimization problem is solved by using the Cobyla (Constrained Optimization BY Linear Approximations) algorithm. To consider rule compliance, the optimization problem is modified to:

$$\begin{aligned} &\text{argmin } \bar{J}(\boldsymbol{x},\boldsymbol{u}) = \frac{J(\boldsymbol{x},\boldsymbol{u})}{N_{\mathrm{p}} B_{\mathrm{OS}}^2} + \varrho_1 \sigma_1 \tau^2 \\ &\text{subject to:} \\ &\mathrm{C1};\mathrm{C2};\mathrm{C3};\mathrm{C4} \\ &\mathrm{C5}: \varrho_1 \varrho_2 \left(\tau - \frac{r_{N_{\mathrm{p}}}}{r_{\max}}\right) \geq 0 \end{aligned} \tag{19}$$

where τ is a slack variable to softly limit the steering direction, σ_1 is the penalty factor for the slack variable, ϱ_1 is a decision parameter for the steering direction constraint, ϱ_2 is a parameter to determine the steering direction, $r_{N_{\mathrm{p}}}$ is the stabilized yaw rate, and $r_{\max}$ is the maximum yaw rate used for normalization. The slack variable becomes an additional optimization variable, and it will converge towards zero to reduce the cost of the objective function. The penalty is only applied when the steering direction constraint (C5) is necessary, which occurs in situations where there is a collision risk, the other ship is dynamic, and the own ship has not yet steered to the requested side:

$$\varrho_1 = \begin{cases} 1, & \text{if the steering direction constraint is needed} \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

It is crucial to cancel the steering direction constraint once the ship has steered to the correct side to minimize deviations from the planned path and to recover the initial voyage. The steering direction constraint is added to the optimization problem, where ϱ_2 is set as:

$$\varrho_2 = \begin{cases} -1, & \text{if steering to starboard side} \\ 1, & \text{if steering to the port side} \\ 0, & \text{otherwise (standing on)} \end{cases} \tag{21}$$

The yaw rate is required to be smaller than τ when the steering direction constraint is set to the port side. Consequently, with a very large penalty factor, the slack variable will be constrained very close to zero, giving preference to negative values for the stabilized yaw rate (steering to port side).

3.6.2 Unconstrained optimization

The constraints C1 to C3 actually all describe the system's dynamic response, which can be included in the predictive model when calculating the predictive ship states in the objective function. The remaining constraints can be eliminated by incorporating penalty functions into the objective function, which is a widely-used technique to convert constrained optimization to unconstrained optimization. The corresponding optimizer is then termed UCobyla (Unconstrained Cobyla) to distinguish it from the aforementioned Cobyla optimizer. The objective function is modified to:

$$\operatorname{argmin} \bar{J}(\boldsymbol{x},\boldsymbol{u}) = \frac{J(\boldsymbol{x},\boldsymbol{u}) + \varrho_1\sigma_2 P_{0,0}\left(-\frac{\varrho_2 r_{N_\mathrm{p}}}{r_{\max}}\right) + \sigma_3 \sum_{j=1}^{N_{\mathrm{ob}}^{\mathrm{G}}}\sum_{i=1}^{N_\mathrm{p}} P_{j,i}\left(\frac{d_{j,i}-D_{j,i}}{D_{j,i}-D_{j,i}^{\mathrm{cm}}}\right)}{N_\mathrm{p}B_{\mathrm{OS}}^2} \tag{22}$$

where $\boldsymbol{x}_{i+1} = f(\boldsymbol{x}_i, \overline{\boldsymbol{u}}_i)$, $\overline{\boldsymbol{u}}_i = \boldsymbol{c}(\boldsymbol{u}_i)$, $\boldsymbol{c}(\boldsymbol{u})$ is a constraint function to filter $\boldsymbol{u}$ with the physical constraints (C2 and C3), $N_{\mathrm{ob}}^{\mathrm{G}}$ is the number of obstacles that the own ship needs to give way to. The penalty function is defined as:

$$P_{j,i}(p) = \begin{cases} 0, & p \geq 0 \\ |p|^m, & \text{otherwise} \end{cases} \tag{23}$$

including $P_{0,0}$. $|p| \leq 1$, σ_2 is the penalty factor for the terminal yaw rate constraint, σ_3 is the penalty factor for the safety distance constraints, and $D_{j,i}^{\mathrm{cm}}$ is the minimum distance to avoid a contour conflict with the *j*-th obstacle at the *i*-th time step. The safety distance considers the contour margins and the additional margins, and it is calculated using Eq. (14). When setting the additional margins to zero in Eq. (14), $D_{j,i}^{\mathrm{cm}}$ can be calculated. In addition, when there is a collision within a prediction horizon, the follow-up penalties will be maintained as the penalty value when the collision begins ($i = i^*$):

$$d_{j,i^*} - D_{j,i^*}^{\mathrm{cm}} \leq 0 \rightarrow P_{j,i}(p) = P_{j,i^*}(p), i = i^* + 1, \ldots, N_{\mathrm{p}} \tag{24}$$

Therefore, the penalty for the safety distance violence reaches the maximum value, σ_3, when $d_{j,i} \leq D_{j,i}^{\mathrm{cm}}$. The penalty for the violence of the yaw rate constraint has the maximum value of σ_2.

4 Experimental tests

4.1 Experimental setup

The Towing Tank for Manoeuvres in Confined Water (cooperation Flanders Hydraulics and Ghent University) facilitates the validation of autonomous navigation applications by providing a controllable environment and enabling fully automated free running tests. A detailed description of the facility can be found in (Delefortrie et al., 2016). Figure 7 illustrates the layout of the towing tank, where the coordinate systems—the earth-fixed coordinate system and the body-fixed coordinate system—are depicted.

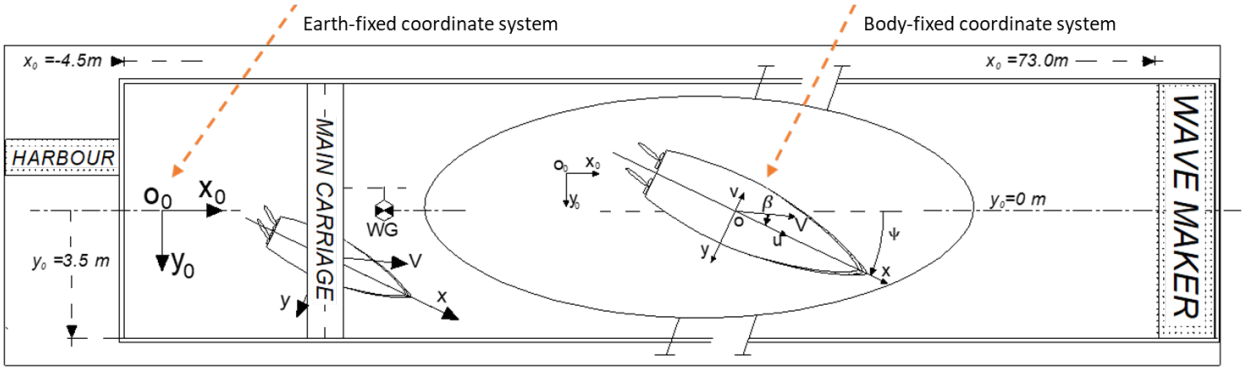


Figure 7. Layout and coordinate systems of the Towing Tank of Manoeuvres in Confined Water

A T01 bulk carrier of 4.136m in length (in model scale) is employed as the control plant. The ship is equipped with a single rudder and a single propeller. Experimental tests are carried out at five different speeds: A (0.184m/s), B (0.307m/s), C (0.4030m/s), E (0.553m/s) and F (0.676m/s), which correspond to 3kn, 5kn, 7kn, 9kn and 11kn in full scale respectively. The under keel clearance is set to 25% ship draft, leading to a shallow water condition. The main characteristics of the T01 ship and a more detailed experimental setting can be referred to (He et al., 2025a).

The controller's performance in each experimental test is evaluated by utilizing the metrics: mean track error integral ($mTEI$), maximum track error (MTE), mean rudder integral (mRI), mean rudder total variation ($mRTV$), minimum contour margin (MCM), and mean violated safety margin integral ($mVSMI$). The definitions of $mTEI$, MTE, mRI and $mRTV$ are introduced in (C. Chen et al., 2020; He et al., 2023). $mTEI$ measures the cross-track deviation from the planned route, while mRI measures the rudder effort made to complete the manoeuvre.

The safety distance considers both contour margins and additional margins, and it is calculated using Eq. (14). When the additional margins are set to zero in Eq. (14), the minimum distance to avoid a contour conflict with the *j*-th obstacle at the *i*-th time step, $D_{j,i}^{\mathrm{cm}}$, can be calculated. Therefore, MCM is calculated as:

$$\begin{aligned} MCM &= \min_i(d_i^{\mathrm{cm}}) \\ d_i^{\mathrm{cm}} &= \min_j\left(d_{j,i} - D_{j,i}^{\mathrm{cm}}\right) \\ d_{j,i} &= \sqrt{\left(x_i - x_{j,i}^{\mathrm{obs}}\right)^2 + \left(y_i - y_{j,i}^{\mathrm{obs}}\right)^2} \end{aligned} \tag{25}$$

If MCM>0, it indicates that there is no collision during the voyage. A larger MCM suggests a more satisfactory safety. The safety margin is calculated when setting additional margins:

$$d_i^{\mathrm{sm}} = \min_j\left(d_{j,i} - D_{j,i}\right) \tag{26}$$

The violated safety margin (margins smaller than 0) is calculated as:

$$d_i^{\mathrm{vsm}} = \min(0, d_i^{\mathrm{sm}}) \tag{27}$$

Therefore, $mVSMI$ is then obtained as:

$$mVSMI = \frac{1}{N_1 - N_0 + 1}\sum_{i=N_0}^{N_1} |d_i^{\mathrm{vsm}}| \tag{28}$$

where N_0 and N_1 are respectively the starting and the end time steps for the evaluation, respectively. $mVSMI$ measures the extent to which the safety distance constraint is violated. The smaller the $mVSMI$,

the less the violation of the safety distance constraint, and a zero $mVSMI$ means that the safety distance constraint is always satisfied during the voyage.

4.2 Collision avoidance with static obstacles

4.2.1 Tests with physical obstacles

Experimental tests were first conducted in the towing tank to avoid collisions with physical obstacles. Two watertight cubic obstacles constructed by concrete plywood were installed in the tank, as shown in Figure 8. The figure demonstrates that the T01 ship interacting with the obstacles and attempting to avoid collisions. These obstacles were ballasted by concrete blocks and remained static during the tests. Obstacle A measures 2.5m×0.7m×0.4m (length×width×height) and is located at (16.75m, -0.85m) in the tank-fixed coordinate system. Obstacle B is twice as long as Obstacle A and is installed at (36.5m, 0.85m). These cubic obstacles generate hydrodynamic interaction forces on the ship when it passes by.

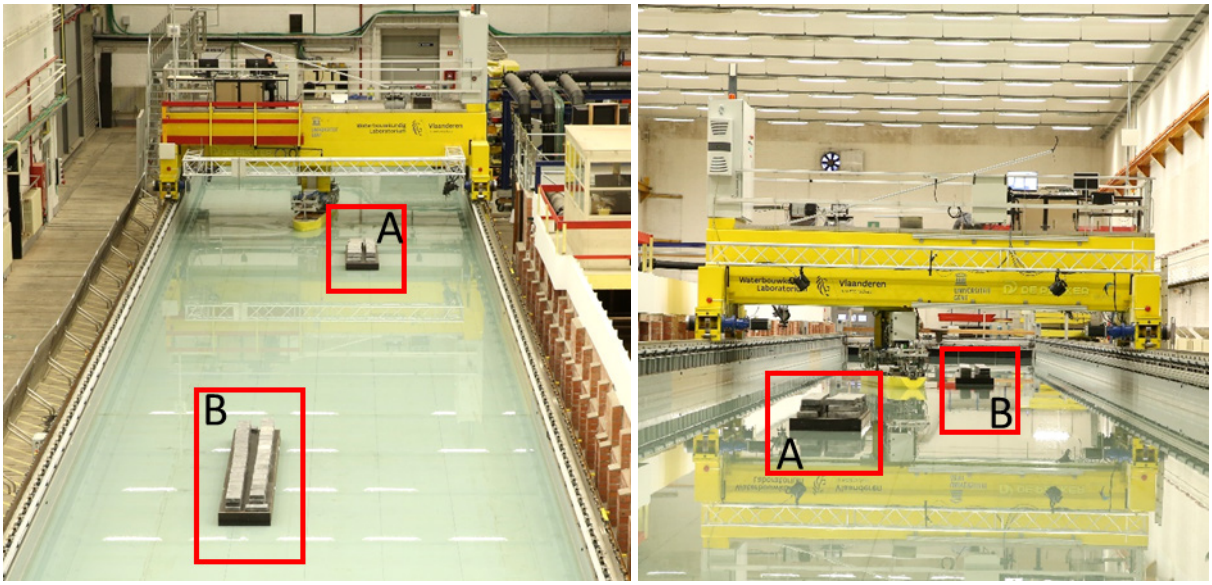


Figure 8. T01 ship interacts with cubic obstacles installed in the tank

The additional lateral and longitudinal margins included in the calculation of the safety distance are set to proportions of the ship's breadth and length, respectively, as:

$$d_{\mathrm{Ba}} = k_{\mathrm{Ba}}B,\ d_{\mathrm{La}} = k_{\mathrm{La}}L \tag{29}$$

where k_{Ba} and k_{La} are the proportional factors called nondimensional additional margins. Three different MPC modes are involved in the testing as listed in Table 4.

Table 4. MPC modes involved in the experimental testing

MPC mode	Predictive model	Speed control
M1	3-DOF simplified model	NO
M2	3-DOF simplified model	YES
M3	Nomoto model	NO

The factors k_{Ba} and k_{La} vary from 0.8 to 1.6 for tests conducted at each speed. However, when the MPC mode is set to M1, an extra variation of 1.8 is added. Figure 9 and Figure 10 respectively show the $mTEI$, MCM, and $mVSMI$ for collision avoidance tests where the nondimensional additional margins and the MPC mode settings were varied. The legend, TA to TF, represents the series of track control tests conducted at the speed class A to F, respectively. A total of 80 tests were planned. Among them, 10 tests failed to finish (stopped by the safety mechanism of the carriage as the ship was too close to the obstacles or the tank walls) and 8 tests were not executed (4 at speed class E, and 4 at speed class F) due to the limited experimental period. No metric values are provided for tests that failed or were not executed.

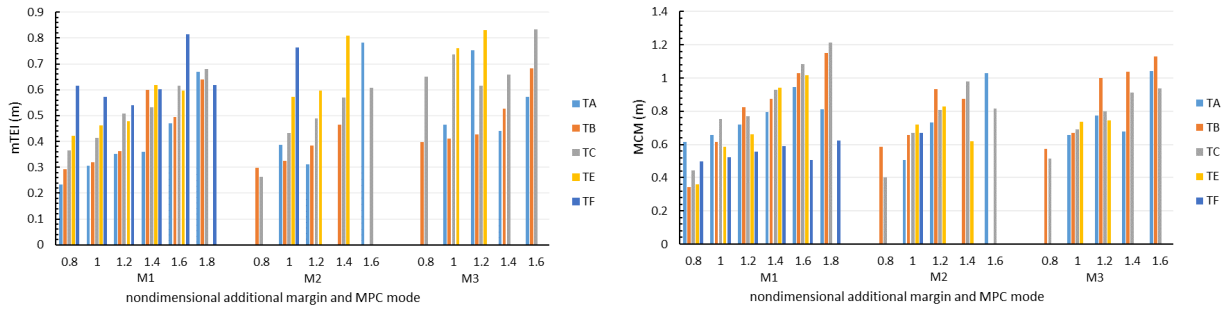
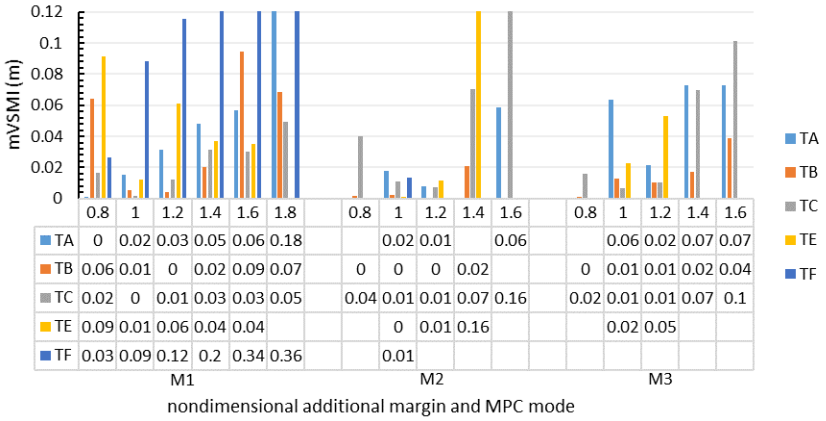


Figure 9. $mTEI$ and MCM for collision avoidance tests with physical obstacles while varying k_{Ba}, k_{La} and MPC mode.

A basic trend observed is that with increased k_{Ba} and k_{La}, both the MCM and $mTEI$ increase, indicating that setting a larger additional margin allows the ship to keep more space from encountering obstacles, thus enhancing safety. However, this leads to greater deviation from the planned route, which can also be visually observed in Figure 11. The profile of the ship is plotted at intervals of 10 seconds for all graphs.



	M1						M2					M3				
	0.8	1	1.2	1.4	1.6	1.8	0.8	1	1.2	1.4	1.6	0.8	1	1.2	1.4	1.6
TA	0	0.02	0.03	0.05	0.06	0.18		0.02	0.01		0.06		0.06	0.02	0.07	0.07
TB	0.06	0.01	0	0.02	0.09	0.07	0	0	0	0.02		0	0.01	0.01	0.02	0.04
TC	0.02	0	0.01	0.03	0.03	0.05	0.04	0.01	0.01	0.07	0.16	0.02	0.01	0.01	0.07	0.1
TE	0.09	0.01	0.06	0.04	0.04			0	0.01	0.16			0.02	0.05		
TF	0.03	0.09	0.12	0.2	0.34	0.36		0.01								

Figure 10. $mVSMI$ for collision avoidance tests with physical obstacles while varying k_{Ba}, k_{La} and MPC mode.

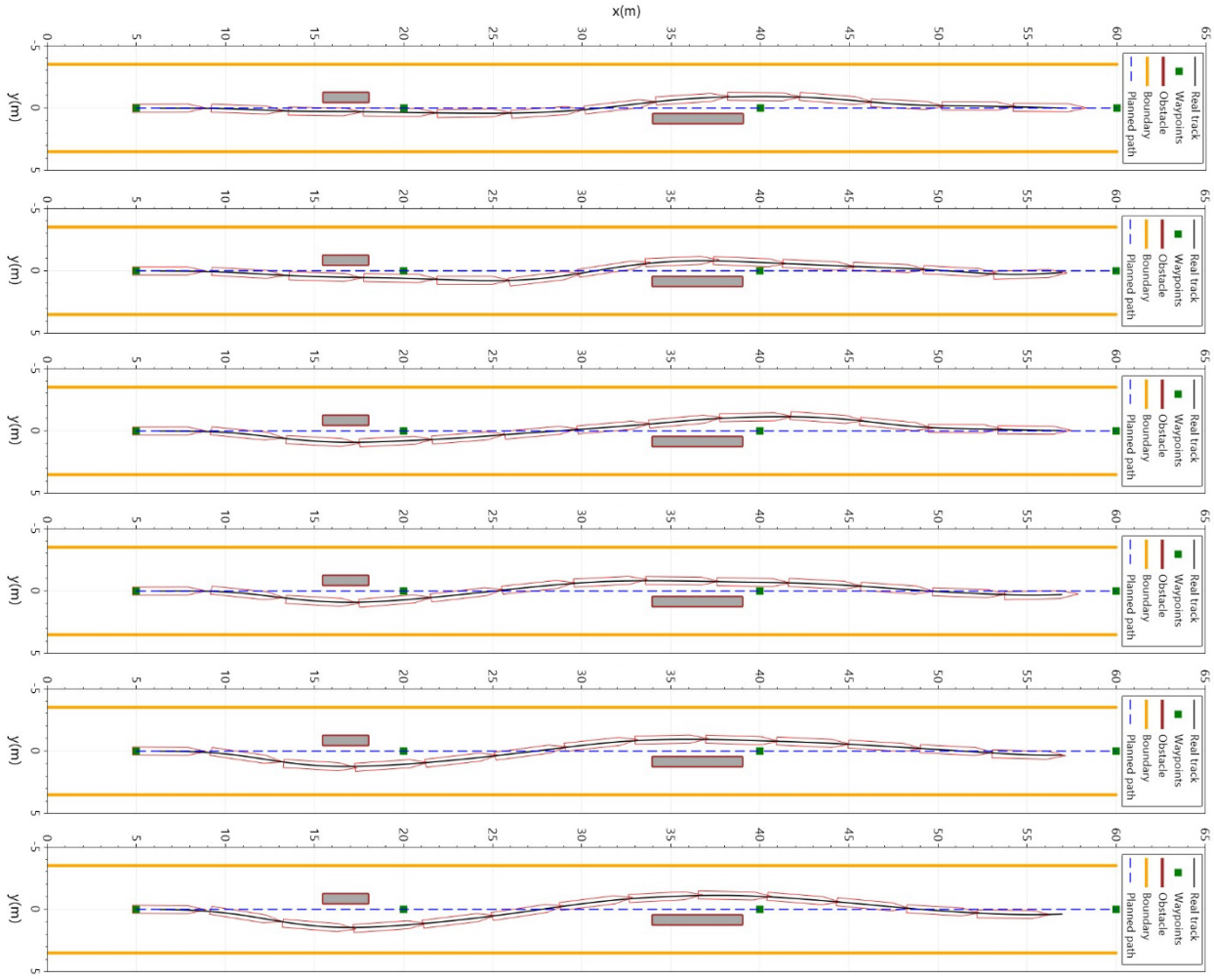


Figure 11. Tracks of collision avoidance tests with physical obstacles at speed class C while $k_{\mathrm{Ba}} = k_{\mathrm{La}} =$ 0.8, 1.0, 1.2, 1.4, 1.6 and 1.8 (top to bottom), MPC mode = M1.

Regarding the speed effect, a higher speed usually leads to a larger $mTEI$ but slightly changes MCM, implying a greater difficulty in achieving collision avoidance at a higher speed. Figure 12 illustrates tracks with the same parameter setting at different speeds, where the deviation from the planned route is seen increasing with speed.

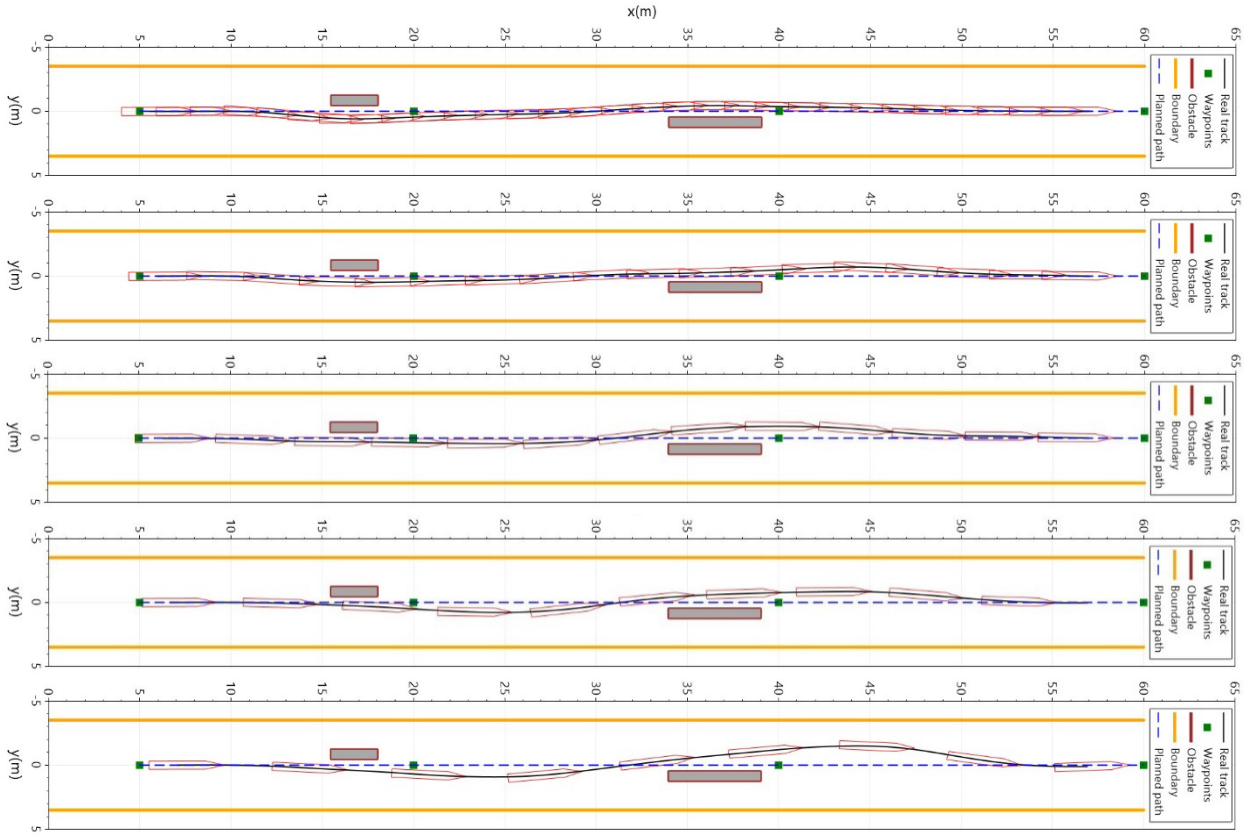


Figure 12. Tracks of collision avoidance tests with physical obstacles at different speed classes (A, B, C, E and F; top to bottom). $k_{\mathrm{Ba}} = k_{\mathrm{La}} = 0.8$, MPC mode = M1, and $K_{\mathrm{OB}} = 2$.

Regarding MPC modes, collision avoidance can be achieved even when employing the Nomoto model as the predictive model (M3). Implementing M1 generally results in a smaller $mTEI$ and a larger MCM, indicating M1 performs best. M3 performs worst according to the metrics $mTEI$ and MCM.

Figure 10 shows the $mVSMI$ for each test, revealing that in most tests the safety distance constraints are not strictly satisfied. Tests with preferable settings are selected to check compliance with the safety distance constraint ($mVSMI$), as shown in Figure 13, where the nondimensional additional margin is limited to 1 and 1.2. When implementing MPC mode M2, the $mVSMI$ is usually the smallest, implying M2 is most reliable.

The is likely because the prediction accuracy of the 3-DOF simplified predictive model is higher than that of the Nomoto model, and mode M2's speed control probably provides more feasible solutions to satisfy the safety distance constraint.

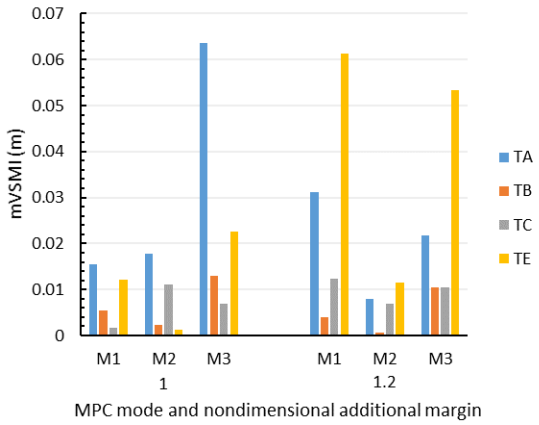


Figure 13. $mVSMI$ for collision avoidance tests with physical obstacles with preferable parameter settings. K_{OB} = 2.

Afterward, a variation of the prediction horizon factor K_{OB} from 2 to 3.5 shows that:

(a) this parameter has a limited effect;

(b) the setting, K_{OB} = 2, demonstrates the best performance according to the matrices $mTEI$ and MCM;

(c) the controller's reliability decreases as the prediction horizon lengthens due to reduced prediction accuracy, according to the $mVSMI$;

(d) with a smaller prediction horizon, the ship takes later but more aggressive reactions to solve conflicts. The smoothness of the real track benefits from a larger prediction horizon.

To summarize, it is recommended to set sufficient additional margins to account for potential violations of safety distance constraints (setting the values slightly larger than the actual needs) and to set the prediction horizon factor between 2 and 3 for good performance (the original prediction horizon should be optimized in the path following or trajectory tracking application).

4.2.2 Influence of ship-obstacle interaction

After conducting collision avoidance tests with physical obstacles in the tank, the physical obstacles are removed and the tests were repeated with virtual obstacles. Figure 14 shows the $mTEI$ and mRI evaluated for these tests. It is observed that more tests succeeded with physical obstacles, and in these cases, larger $mTEI$ and mRI were recorded. It seems that the ship-obstacle interaction positively impacts collision avoidance with static obstacles.

Figure 15 shows the tracks, rudder angles and margins of the collision avoidance tests conducted at speed class C. It is seen that the ship failed to return to the original route after avoiding collision with the second obstacle when physical obstacles were not present. This is because the rudder got stuck at the maximum deflection angle for a while, severely violating the safety distance constraint during this period and causing a controller instability. When the bow approached and the stern passed the physical obstacles, they were attracted to the obstacles due to hydrodynamic interactional forces, helping the ship to correct its heading parallel to the obstacles. Therefore, this may positively influence control performance during collision avoidance.

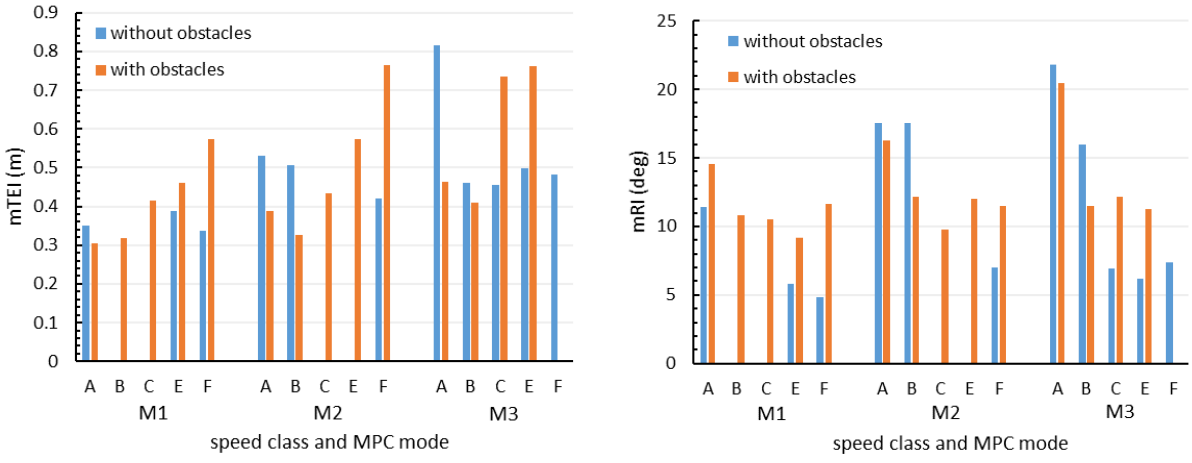


Figure 14. $mTEI$ and mRI of collision avoidance tests conducted with and without physical obstacles.

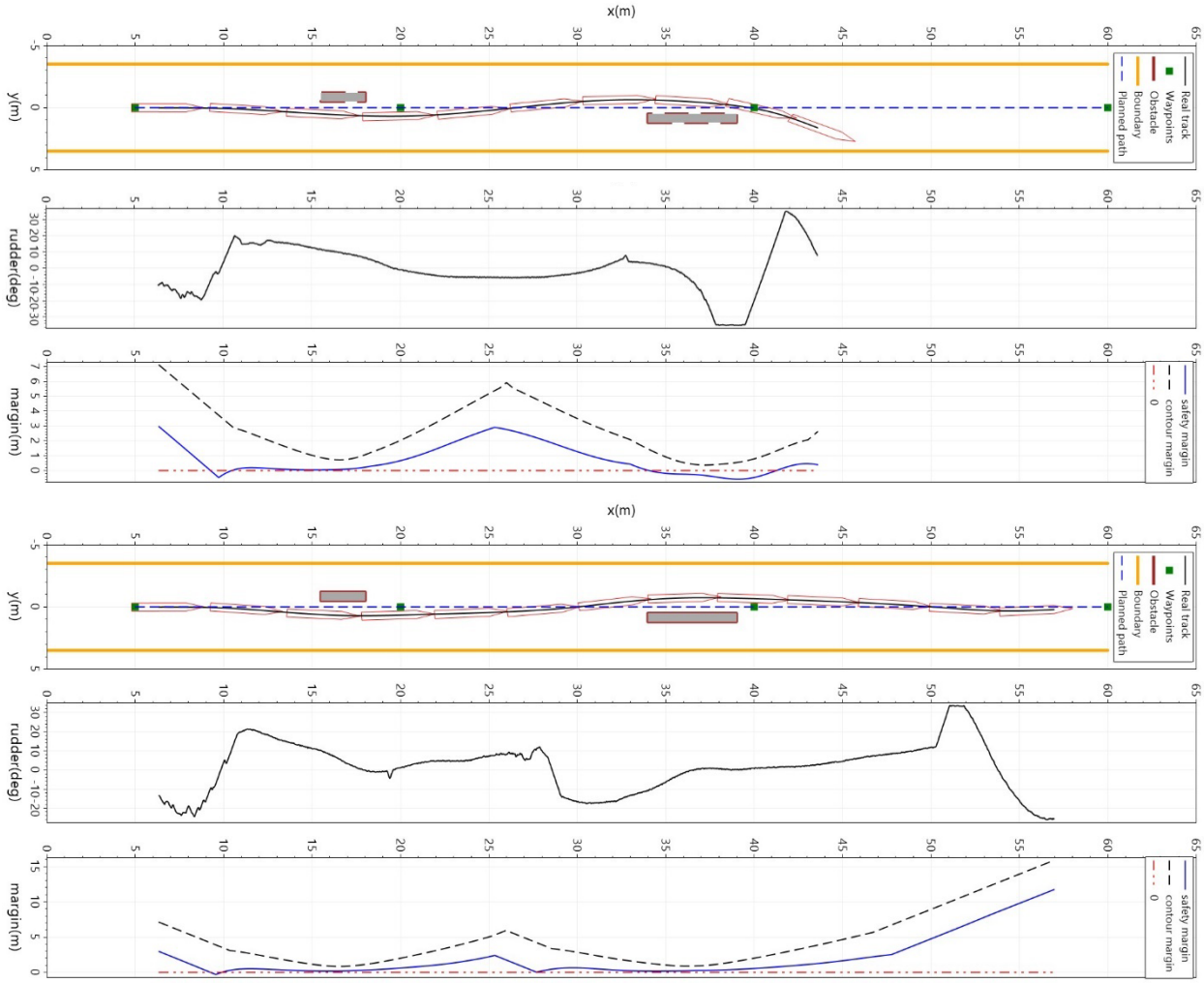


Figure 15. Collision avoidance tests conducted with and without physical obstacles at speed class C. MPC mode = M1. Top three subfigures: without physical obstacles.

4.2.3 Unconstrained optimization

It is suspected that the violation of the safety distance constraints will bring instability to the controller. To avoid this issue, these constraints can be dropped by the optimizer UCobyla. Table 5 lists the control performance metrics for collision avoidance tests with the implementation of the UCobyla optimizer and the Cobyla optimizer, respectively, where speed class = C, MPC mode = M1, $k_{\mathrm{Ba}} = k_{\mathrm{La}} = 1.2$, and $\sigma_3 = 10$. No values are given for failed tests.

The results demonstrate that with the implementation of the UCobyla, the controller is more robust to the setting of K_{OB} as more tests can be completed. When using UCobyla, the track deviation is reduced at the

expense of satisfying of the safety distance constraint. Figure 16 illustrates the tracks of collision tests at speed class C when respectively using the two optimizers. It is evident that the track is better controlled with UCobyla.

Table 5. Control performance metrics for collision avoidance tests with different optimizers, conducted without presence of physical obstacles.

Optimizer	K_{ob}	$mTEI$ [m]	MTE [m]	mRI [deg]	MCM [m]	$mVSMI$ [m]
Cobyla	2					
	2.5					
	3	0.6620	1.8869	9.6264	0.6908	0.0630
	3.5	0.4343	0.8807	6.9797	0.6579	0.0907
UCobyla	2	0.3641	0.8104	6.8590	0.7635	0.0737
	2.5	0.3709	0.6226	5.3069	0.6613	0.1574
	3	0.6174	1.4016	9.5877	0.5836	0.2584
	3.5	0.3678	0.6841	5.8906	0.5272	0.2959

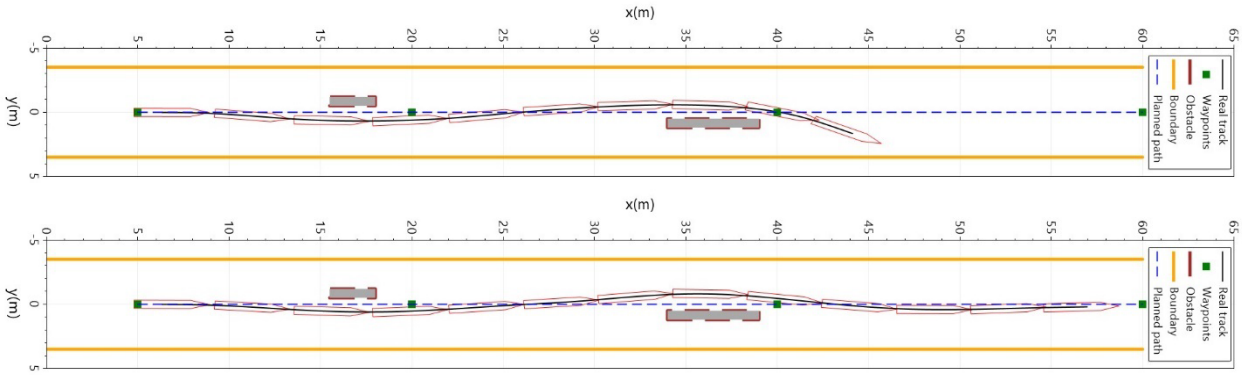


Figure 16. Collision avoidance tests conducted without physical obstacles at speed class C. K_{OB} = 2. Top subfigure: Optimizer = Cobyla, bottom subfigure: Optimizer = UCobyla.

4.2.4 Decrease of longitudinal additional margin

The same nondimensional additional margin was utilized for the tests introduced in the previous sections, but in fact the additional margins around the ship were not equal (the longitudinal margin was larger than the lateral margin) as the ship length and width were used for normalizing the longitudinal and lateral margins respectively. According to the test results, it seems that the safety distance constraints were usually

violated in the longitudinal direction, implying that the requested longitudinal margin may have been too conservative. Therefore, the same additional margin was then set around the ship:

$$k_{\mathrm{La}} = \frac{k_{\mathrm{Ba}} B^{\mathrm{OS}}}{L^{\mathrm{OS}}} \tag{30}$$

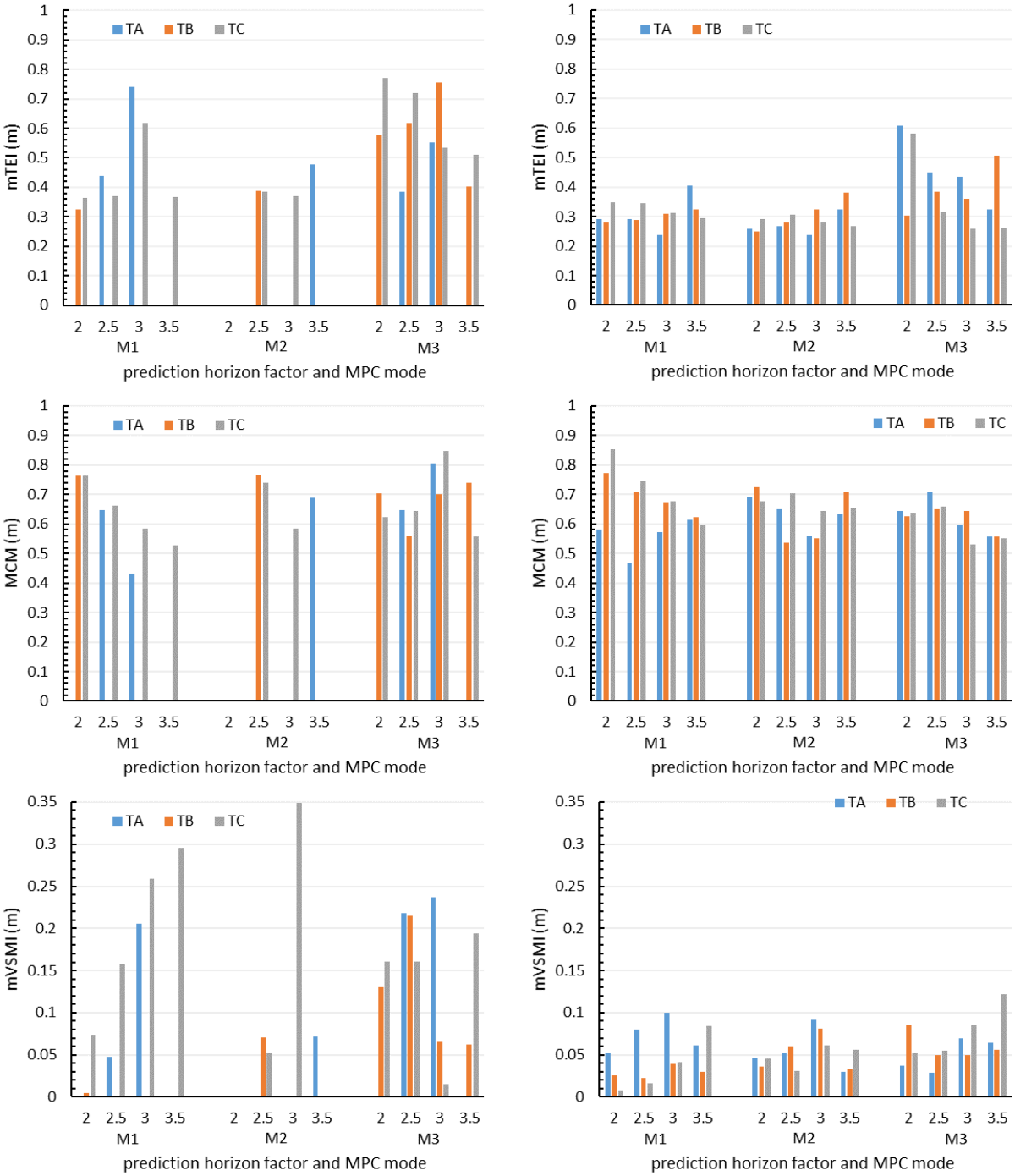


Figure 17. Comparison of $mTEI$, MCM and $mVSMI$ of collision avoidance tests without physical obstacles when applying different additional margin setting strategies. Left column: same nondimensional additional margins, $k_{\mathrm{Ba}} = k_{\mathrm{La}} = 1.2$; Right column: same additional margins, $k_{\mathrm{Ba}} = 1.2$, $k_{\mathrm{La}} = 1.2B^{\mathrm{OS}}/L^{\mathrm{OS}}$.

Experimental tests were repeated to investigate the influence, and test results as shown in Figure 17 indicate that the success rate was significantly increased to 100%. Additionally, $mTEI$, $mVSMI$ and MCM were all reduced.

4.3 Collision avoidance with a moving obstacle

4.3.1 Setup and scenario description

Experimental tests were conducted in an empty tank (physical obstacles removed) to validate the algorithm for collision avoidance with virtually moving obstacles. Since the auxiliary carriage in the towing tank was not available for utilization to captively move another ship, a dynamic obstacle was simulated within the controller to create head-on or overtaking scenarios. Consequently, there were no hydrodynamic ship-ship interactions involved in these tests. Although resolving conflicts with multiple obstacles was considered when designing the controller, the corresponding tests were not conducted due to the limited space of the towing tank. In addition, because of the limited width of the tank, only the S01 and S09 scenarios (as seen in Figure 6) were created. The obstacle simulated had a length of 5m and a width of 0.7m.

For the head-on scenarios, the simulated obstacle approached from the end of towing tank with a course of 180 degrees. θ_0, used to distinguish scenarios in Table 2, is set to 10 degrees. The planned route of the own ship was a straight line along $y_0 = -1$ to provide more starboard space, as the own ship was expected to steer to the starboard side to give way. The speed classes of the own ship and the obstacle (target ship) varied from A to F, generating 25 (=5×5) head-on cases.

For the overtaking scenarios, the simulated obstacle moved ahead of the own ship in the same direction and had a slower speed than the own ship. The speed class of the own ship varied from B to F, while the speed class of the obstacle varied from A to E, generating 10 (= 4 + 3 + 2 + 1) overtaking cases. The planned route

of the own ship was a straight line along $y_0 = 1$ to provide more port side space, as the own ship was expected to steer to the port side to give way.

In each scenario, the course and the speed of the obstacle is kept as constant. Although, in head-on scenarios, the obstacle should also give way to the own ship, its course will not be altered since it is not controlled. The MPC mode is set to M2, with parameters k_{Ba} = 0.8, k_{La} = $0.8B^{\mathrm{OS}}/L^{\mathrm{OS}}$, and K_{OB} = 3.

4.3.2 Test result by using UCobyla

The parameters in the objective function were initially set to $\sigma_2 = 1000$, $\sigma_3 = 100$. Figure 18 presents the values of the control performance indices, MCM, $mVSMI$, MTE and $mTEI$, of the 25 head-on tests. According to the MCM, all head-on tests were successfully completed without the own ship colliding with the target ship (MCM>0). Lower $mVSMI$ values suggest a better compliance with the requested safety margins, with noticeable variations across the different speed classes. Notably, when the target ship was in speed class B, the safety margins were mostly violated. For the MTE and $mTEI$ indices, a general trend was observed: they increased with the own ship's speed.

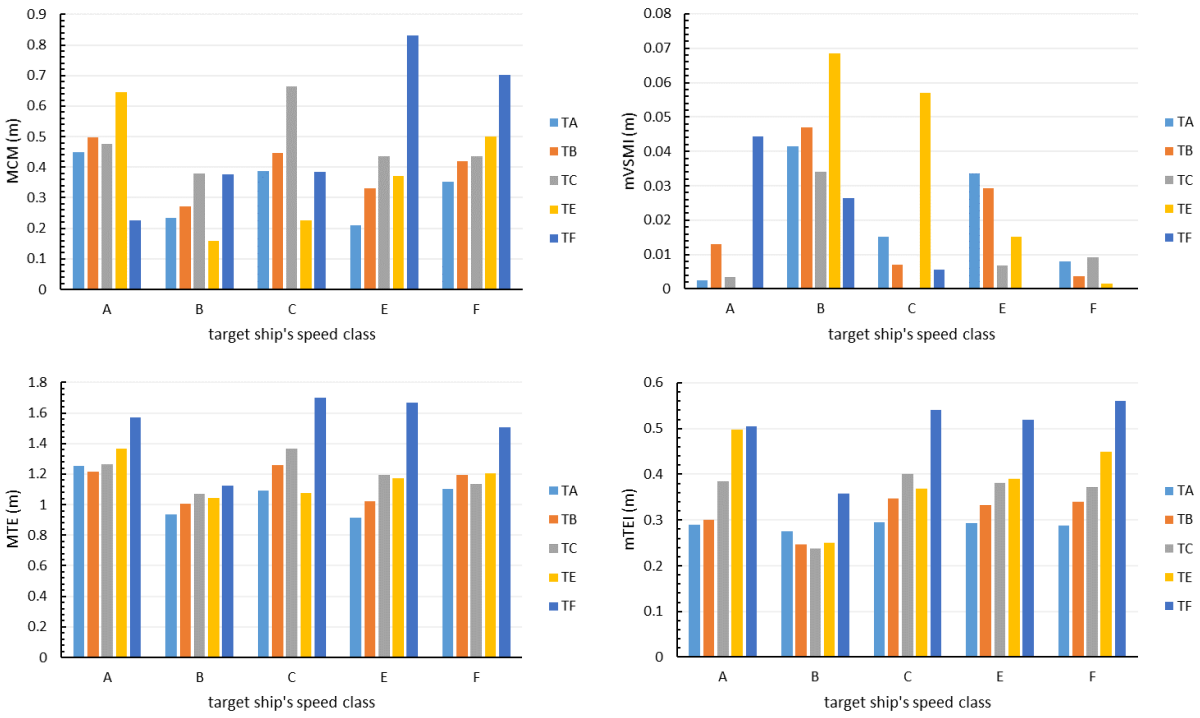


Figure 18. MCM, $mVSMI$, MTE and $mTEI$ of head-on collision avoidance tests. The speed of the target ship and the own ship were variated. Optimizer = UCobyla.

Figure 19 to Figure 20 illustrate the tracks of some head-on tests, where the asterisk symbols indicate the x-position of the middle of the two ships' CPAs (see in Section 1). According to the COLREGs, the own ship should take the starboard side to give way to the target ship. However, when the target ship was in speed class B and the own ship was in speed classes B, C, E and F, the own ship took the wrong side to give way, even though the collision avoidance was successfully achieved. An explanation will be provided later.

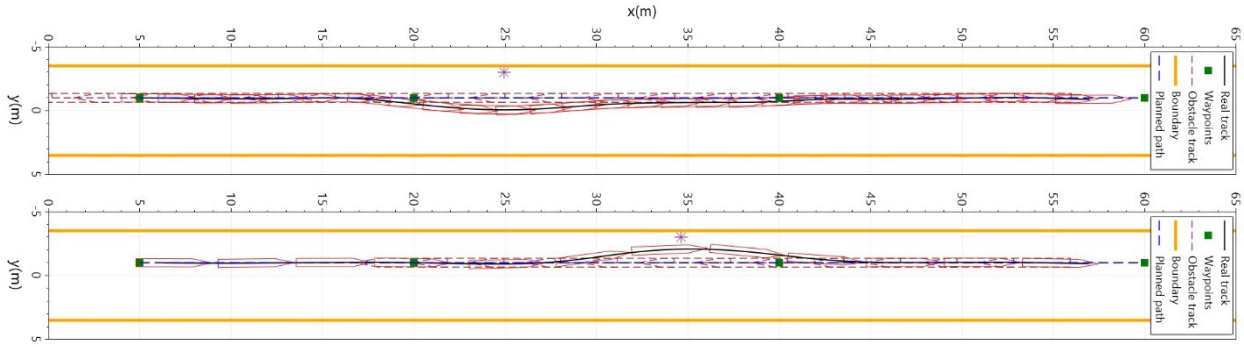


Figure 19. Trajectories of head-on tests when the target ship is in speed class B. From the top to the bottom, each subfigure respectively presents the result when the own ship's speed class varies from A and C.

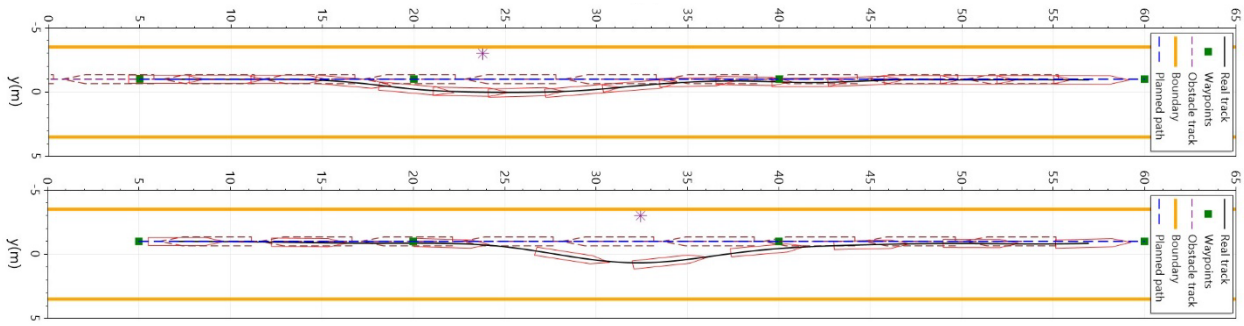


Figure 20. Trajectories of head-on tests when the target ship is in speed class E. From the top to the bottom, each subfigure respectively presents the result when the own ship's speed class varies from B and F.

Table 6 lists the values of the control performance indices for the 10 overtaking tests. 2 tests, T0108A13_TC17100 and T0108A13_TF17300, failed to complete as the own ship incorrectly took the starboard side to give way, where they was less space. In the tests T0108A13_TC17200 and T0108A13_TF17200, the own ship also took the wrong side to give way. Although the two tests were completed, the own ship collided with the obstacle, as indicated by the negative values of MCM. In the

remaining tests, the own ship took the correct side to give way, and the collisions were successfully avoided.

Figure 21 to Figure 22 illustrate the tracks of some overtaking tests.

Table 6. Values of the control performance indices for the overtaking tests. Parameter setting: Optimizer = UCobyla, σ_2 = 1000, σ_3 = 100. Notation: TS = target ship, PS = port side, SB = starboard side.

Test	TS's speed class	$mTEI$ [m]	MTE [m]	MCM [m]	$mVSMI$ [m]	Give-way
T0108A13_TB17100	A	0.569	1.394	0.616	0.073	PS
T0108A13_TC17100	A	-	-	-	-	SB
T0108A13_TE17100	A	0.621	1.875	0.985	0.002	PS
T0108A13_TF17100	A	0.839	2.586	1.594	0.000	PS
T0108A13_TC17200	B	0.334	1.029	-0.026	0.295	SB
T0108A13_TE17200	B	0.876	2.412	0.906	0.100	PS
T0108A13_TF17200	B	0.572	1.263	-2.990	0.623	SB
T0108A13_TE17300	C	0.162	1.309	0.073	0.054	PS
T0108A13_TF17300	C	-	-	-	-	SB
T0108A13_TF17400	E	0.585	1.740	0.097	0.229	PS

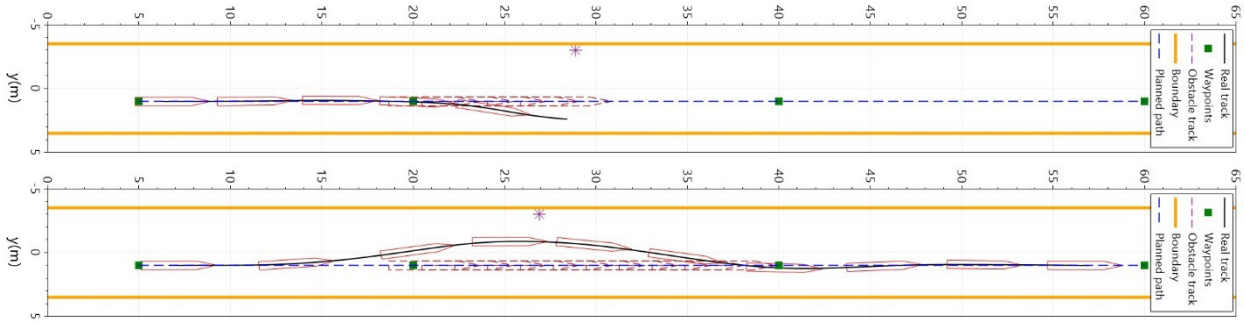


Figure 21. Trajectories of overtaking tests with the target ship's speed at class A. Top to the bottom: the own ship's speed class varies from C to E.

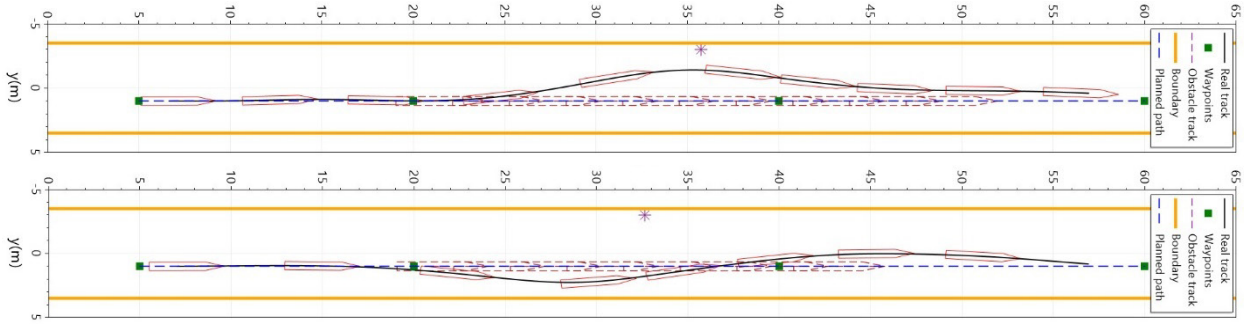


Figure 22. Trajectories of overtaking tests with the target ship's speed at class B. Top to the bottom: the own ship's speed class varies from E to F.

The previous result indicates that the rules set for collision avoidance with another moving ship were not fully respected, which implies that the penalty on the violation of the yaw rate constraint might be insufficient. From the perspective of nondimensionalization, Eq. (22) should be modified to:

$$\bar{J}(\boldsymbol{x},\boldsymbol{u}) = \frac{J(\boldsymbol{x},\boldsymbol{u})}{N_{\mathrm{p}}B_{\mathrm{OS}}^2} + \varrho_1\bar{\sigma}_2 P_{0,0}\left(-\frac{\varrho_2 r_{N_{\mathrm{p}}}}{r_{\max}}\right) + \frac{\bar{\sigma}_3}{N_{\mathrm{p}}}\sum_{j=1}^{N_{\mathrm{ob}}^G}\sum_{i=1}^{N_{\mathrm{p}}} P_{j,i}\left(\frac{d_{j,i}-D_{j,i}}{D_{j,i}-D_{j,i}^{\mathrm{cm}}}\right) \tag{31}$$

where $\bar{\sigma}_2 = \frac{\sigma_2}{N_{\mathrm{p}}B_{\mathrm{OS}}^2}$, and $\bar{\sigma}_3 = \frac{\sigma_3}{B_{\mathrm{OS}}^2}$. With this modification, the penalties are normalized and averaged, allowing the magnitudes of the state error and the penalties for the yaw rate constraint and the safety distance constraint to be respectively reflected by 1, $\bar{\sigma}_2$ and $\bar{\sigma}_3$. With the initial parameter setting, $\bar{\sigma}_3$ = 242 and $\bar{\sigma}_2 \approx$ 5 ($\bar{\sigma}_2$ varies with the own ship's speed due to changes in the prediction horizon). Therefore, the penalty factor σ_2 was too small. σ_2 was then increased to 10^4, and the failed tests were repeated.

Table 7 lists the values of the control performance indices for these repeated tests, showing that all tests were correctly completed and collisions were avoided. Therefore, by increasing the penalty factor σ_2, the rules for collision avoidance can be more effectively respected. Figure 23 to Figure 25 illustrate the tracks for these tests.

Table 7. Values of the control performance indices for the head-on and overtaking tests. Parameter setting: $\sigma_2 = 10^4$, $\sigma_3 = 100$.

Test name	TS's speed class	Scenario	$mTEI$ [m]	MTE [m]	MCM [m]	$mVSMI$ [m]	Give-way
T0108A13_TB18500	B	head-on	0.336	1.207	0.452	0.003	SB
T0108A13_TC18500	B	head-on	0.424	1.330	0.603	0.000	SB
T0108A13_TE18500	B	head-on	0.376	1.158	0.347	0.025	SB
T0108A13_TF18500	B	head-on	0.524	1.652	0.254	0.025	SB
T0108A13_TC18900	A	overtaking	0.591	1.871	0.879	0.015	PS
T0108A13_TC19000	B	overtaking	0.369	1.624	0.192	0.062	PS

T0108A13_TF19000	B	overtaking	0.713	1.275	0.304	0.206	PS
T0108A13_TF19100	C	overtaking	0.938	3.033	0.299	0.078	PS

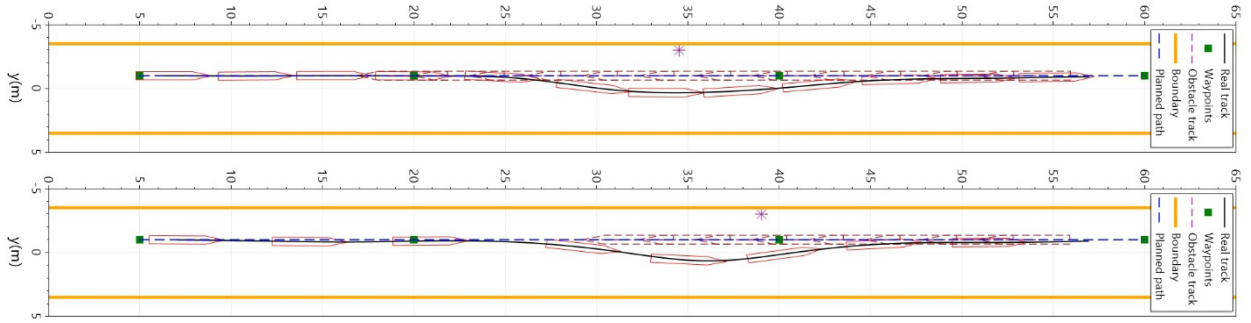


Figure 23. Trajectories of head-on tests with a larger σ_2 and the target ship's speed at class B. Top to the bottom: the own ship's speed class varies from C and F.

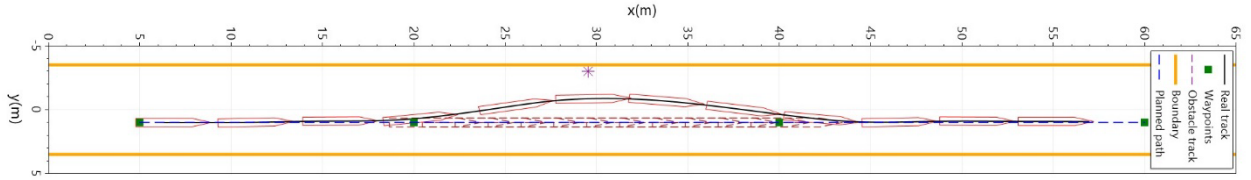


Figure 24. Trajectories of overtaking tests with a larger σ_2 with the target ship's speed at class A and the own ship's speed at class C.

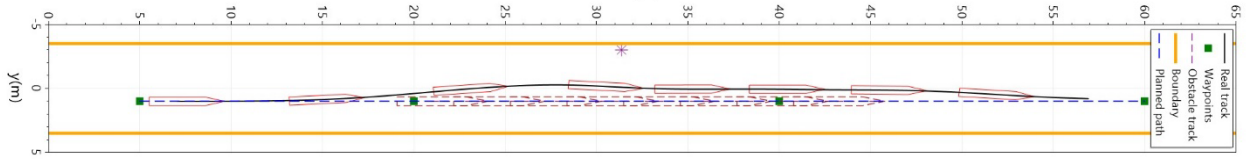


Figure 25. Trajectories of overtaking tests with a larger σ_2 with the target ship's speed at class B and the own ship's speed at class F.

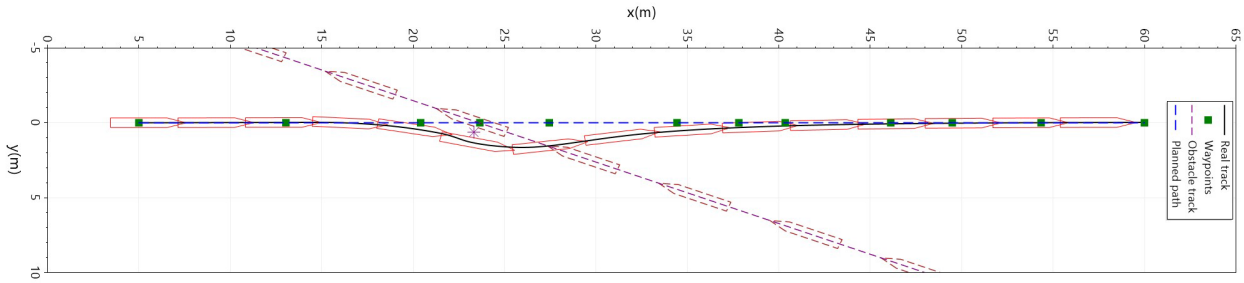
5 Simulation tests

Figure 6 defines all possible collision avoidance scenarios, S01 to S16, for two ships. In the towing tank, only S01 and S09 were tested due to space limitations. However, with simulations, additional scenarios can be explored. It is particularly interesting to test the scenarios where the own ship must give way rather than stand on (maintain her speed and course). Therefore, the remaining scenarios to be tested include S02 and S10 (overtaking) as well as S12, S13 and S14 (crossing). Two Mexico ship models with the same dimensions whose mathematical models are obtained through captive tests in the towing tank are employed, and the

tests are simulated in the environment of the Towing Tank for Manoeuvres in Confined Water (same configuration files for MPC and test conditions are used), extending the towing tank tests.

Ship-ship interaction forces are included in the mathematical model of the simulator. The parameter setting for MPC is: optimizer = UCobyla, $\sigma_2 = 10^4$, $\sigma_3 = 1000$, $m = 0.1$, $k_{\mathrm{Ba}} = 0.8$, $k_{\mathrm{La}} = 0.8B^{\mathrm{OS}}/L^{\mathrm{OS}}$, $K_{\mathrm{OB}} = 3$. Each test is designated by the code T*0*S**. The first two digits, T*, ranges from TA to TF (with A through F representing the speed class of the own ship), the subsequent two digits, 0*, indicate the speed class of the other ship (with 01 corresponding to A, 02 to B, 03 to C, 04 to E, 05 to F), and the last three digits signify the specific collision scenario as illustrated in Figure 6.

In each scenario, a case is illustrated in Figure 26. The symbol '*' on the track plot represents the midpoint of the two ships' CPA (the closest point of approach), which indicates there was adequate space when the two ships passed each other. This is further confirmed by $MCM > 0$ in Table 8. Although the safety distance constraints were slightly violated, the contour margin was successfully maintained. Meanwhile, all evasive manoeuvres were executed on the correct passing side, ensuring adherence to protocol. It should be noticed that the giving-way in overtaking scenario S10 led to crossing the target ship's bow, which might be risky. In fact, the steering direction in overtaking is not restricted in COLREGs, but it is requested to steer to the portside in inland waterways in this study as explained in Section 3.1. These scenarios may not happen in inland waterways since a lot of space is required. The steering constraints should be released or modified to starboard at sea to find better solutions.



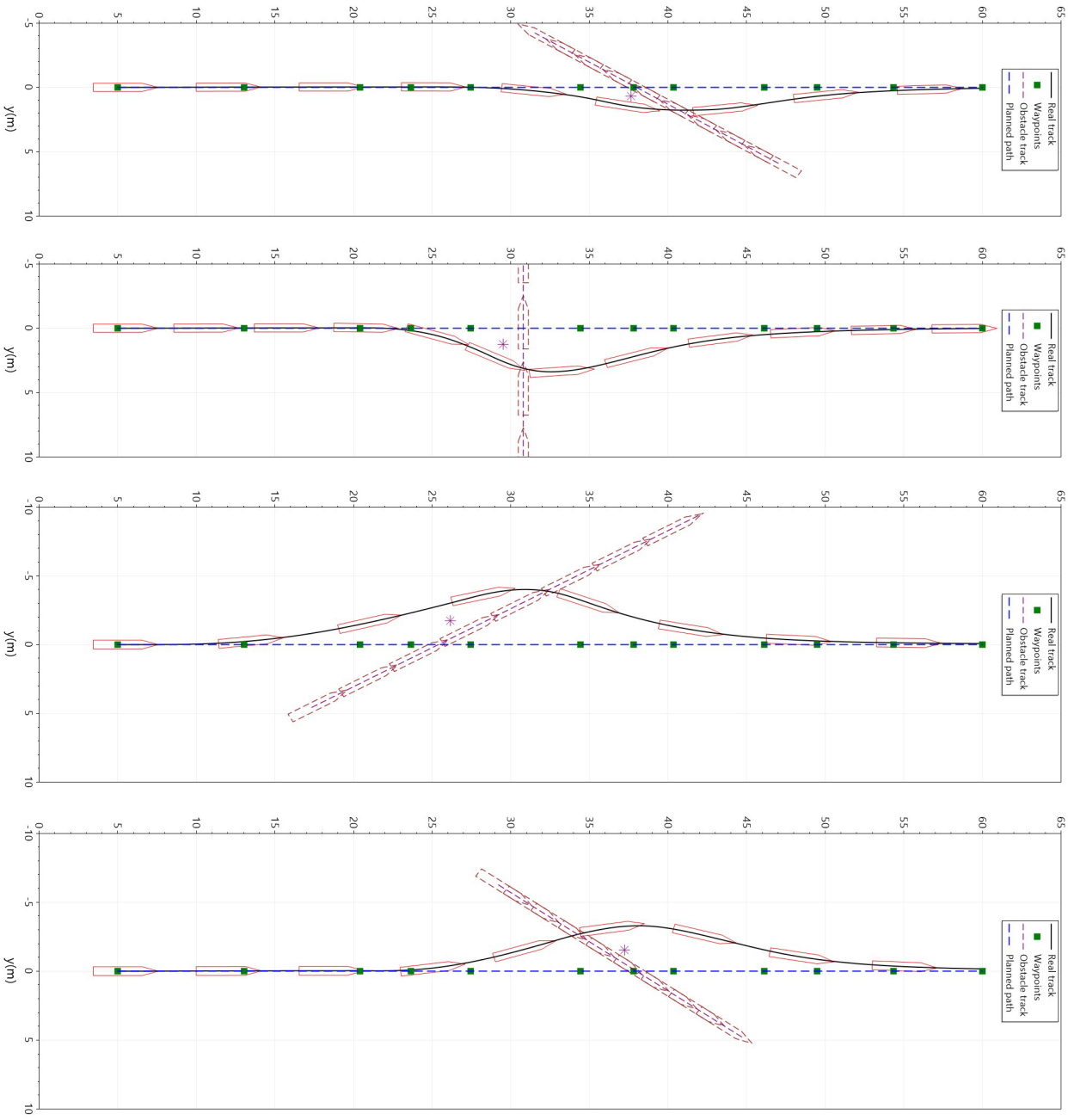


Figure 26 Collision avoidance tests TB04S02, TE01S13, TC03S14, TF03S10, TE01S12

Table 8 Control performance indexes of collision avoidance tests TB04S02, TE01S13, TC03S14, TF03S10, TE01S12

Test	$mTEI$ [m]	MTE [m]	mRI [deg]	$mSEI$ [m/s]	MCM [m]	$mVSMI$ [m]	Give-way
TB04S02	0.38	1.83	4.52	0.01	0.45	0.005	SB
TE01S13	0.48	1.77	3.38	0.01	0.12	0.019	SB
TC03S14	0.86	3.38	8.60	0.02	0.19	0.005	SB
TF03S10	1.55	6.65	2.53	0.05	1.18	0.000	PS
TE01S12	1.02	3.29	4.62	0.02	0.80	0.000	PS

6 Conclusion

This research presents an MPC based collision avoidance mechanism for autonomous ships, which can be applied in solving conflicts with both static and dynamic obstacles. Comprehensive free running experimental tests conducted on a scale model ship in a confined towing tank have led to several key findings and observations:

a) Collision avoidance controller: It effectively resolves conflicts with both static and moving obstacles in a confined environment. The margin to obstacles can be controlled by adjusting the additional margin parameters. It is advisable to avoid setting a significantly larger addition margin in the longitudinal direction and instead set uniform additional margins around the own ship. The prediction horizon factor K_{OB} should be set between 2 and 3 to balance the prediction accuracy and the timeliness of control actions. A 3-DOF predictive model enhances the control performance, and enabling speed control provides better solutions.

b) Optimizers: For resolving conflicts with static obstacles, both Cobyla and UCobyla optimizers delivered very high success rates. The Cobyla optimizer ensured better compliance with safety distance constraints but lacked robustness to provide expected solutions in dynamic obstacles scenarios. Particularly, violating the constraints caused instability in the controller. Future improvements can consider using hard constraints for the stabilizing yaw rate instead of soft constraints or designing alternative constraints for restraining the steering direction. The UCobyla optimizer, which converts constrained optimization to unconstrained optimization, improved the controller robustness and increased the success rate of collision avoidance. It reduced the track deviation at a cost of violating the safety distance constraints more. By increasing the penalty factor σ_2, steering rules were adhered to more effectively. Future research should explore increasing the penalty factor σ_3 to check if compliance with safety distance constraints can be improved.

c) The relevant differences among the hydrodynamic effect of using either physical or virtual static obstacles for the design of control algorithms have been shown for the tested scenarios.

For general future improvement, incorporating distance constraints to borders (banks or waterway borders defined at the level of the minimum under keel clearance) in the MPC objective function could enhance the safety in inland navigation.

CRediT authorship contribution statement

Hongwei He: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **José Villagómez Rosales**: Conceptualization, Formal analysis, Investigation, Resources, Data curation, Writing - review & editing, Supervision, Project administration. **Thibaut Van Zwijnsvoorde**: Conceptualization, Formal analysis, Investigation, Resources, Data curation, Writing - review & editing, Supervision, Project administration. **Evert Lataire**: Conceptualization, Resources, Data curation, Writing - review & editing, Supervision, Project administration. **Guillaume Delefortrie**: Conceptualization, Methodology, Software, Validation, Investigation, Resources, Writing - review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Reference

Abdelaal, M., Fränzle, M., Hahn, A., 2018. Nonlinear model predictive control for trajectory tracking and collision avoidance of underactuated vessels with disturbances. Ocean Engineering 160, 168–180. https://doi.org/10.1016/j.oceaneng.2018.04.026

Brcko, T., Androjna, A., Srše, J., Boć, R., 2021. Vessel multi-parametric collision avoidance decision model: Fuzzy approach. J Mar Sci Eng 9, 1–24. https://doi.org/10.3390/jmse9010049

Cao, S., Fan, P., Yan, T., Xie, C., Deng, J., Xu, F., Shu, Y., 2022. Inland Waterway Ship Path Planning Based on Improved RRT Algorithm. J Mar Sci Eng 10. https://doi.org/10.3390/jmse10101460

Chen, C., Delefortrie, G., Lataire, E., 2020. Experimental investigation of practical autopilots for maritime autonomous surface ships in shallow water. Ocean Engineering 218, 108246. https://doi.org/10.1016/j.oceaneng.2020.108246

Chen, P., Huang, Y., Papadimitriou, E., Mou, J., van Gelder, P., 2020. Global path planning for autonomous ship: A hybrid approach of Fast Marching Square and velocity obstacles methods. Ocean Engineering 214, 107793. https://doi.org/https://doi.org/10.1016/j.oceaneng.2020.107793

Chiang, H.-T.L., Tapia, L., 2018. COLREG-RRT: An RRT-Based COLREGS-Compliant Motion Planner for Surface Vehicle Navigation. IEEE Robot Autom Lett 3, 2024–2031. https://doi.org/10.1109/LRA.2018.2801881

Delefortrie, G., Geerts, S., Vantorre, M., 2016. The towing tank for manoeuvres in shallow water, in: Proceedings of the 4th MASHCON, 2016, Hamburg, Germany. pp. 226–235. https://doi.org/10.18451/978-3-939230-38-0

Du, L., Banda, O.A.V., Huang, Y., Goerlandt, F., Kujala, P., Zhang, W., 2021. An empirical ship domain based on evasive maneuver and perceived collision risk. Reliab Eng Syst Saf 213, 107752. https://doi.org/10.1016/j.ress.2021.107752

Eriksen, B.-O.H., Breivik, M., 2017. MPC-Based mid-level collision avoidance for asvs using nonlinear programming, in: 2017 IEEE Conference on Control Technology and Applications (CCTA). pp. 766–772. https://doi.org/10.1109/CCTA.2017.8062554

Fujii, Y., Tanaka, K., 1971. Traffic capacity. The Journal of navigation 24, 543–552.

García Maza, J.A., Argüelles, R.P., 2022. COLREGs and their application in collision avoidance algorithms: A critical analysis. Ocean Engineering 261, 112029. https://doi.org/10.1016/j.oceaneng.2022.112029

Hagen, I.B., Kufoalor, D.K.M., Brekke, E.F., Johansen, T.A., 2018. MPC-based Collision Avoidance Strategy for Existing Marine Vessel Guidance Systems, in: 2018 IEEE International Conference on Robotics and Automation (ICRA). pp. 7618–7623. https://doi.org/10.1109/ICRA.2018.8463182

He, H., Mansuy, M., Verwilligen, J., Delefortrie, G., Lataire, E., 2024. Global path planning for inland vessels based on fast marching algorithm. Ocean Engineering 312, 119172. https://doi.org/10.1016/j.oceaneng.2024.119172

He, H., Van Zwijnsvoorde, T., Lataire, E., Delefortrie, G., 2023. Model predictive controller for path following ships validated by experimental model tests. Ocean Engineering 288, 115971. https://doi.org/10.1016/j.oceaneng.2023.115971

He, H., Van Zwijnsvoorde, T., Villagómez Rosales, J., Lataire, E., Delefortrie, G., 2025a. Comprehensive experimental testing of parameter sensitivity on a model predictive controller for path following in

shallow water. Ocean Engineering 326, 120882. https://doi.org/https://doi.org/10.1016/j.oceaneng.2025.120882

He, H., Villagómez Rosales, J., Van Zwijnsvoorde, T., Lataire, E., Delefortrie, G., 2025b. Experimental assessment of speed adaptive track control of rudder-propeller-actuated ships based on model predictive control. Ocean Engineering 326. https://doi.org/10.1016/j.oceaneng.2025.120824

IMO, 1972. Convention on the International Regulations for Preventing Collisions at Sea, 1972 (COLREGs) [WWW Document]. URL https://www.imo.org/en/About/Conventions/Pages/COLREG.aspx (accessed 5.20.24).

KEARON, J., 1977. Computer programs for collision avoidance and traffic keeping, in: Conference on Mathematical Aspects on Marine Traffic. London: Academic Press.

Langxiong, G., Li, X., Yan, T., Song, L., Xiao, J., Shu, Y., 2025. Intelligent ship path planning based on improved artificial potential field in narrow inland waterways. Ocean Engineering 317, 119928. https://doi.org/https://doi.org/10.1016/j.oceaneng.2024.119928

Lazarowska, A., 2015. Ship's Trajectory Planning for Collision Avoidance at Sea Based on Ant Colony Optimisation. Journal of Navigation 68, 291–307. https://doi.org/10.1017/S0373463314000708

Liu, Y., Liu, W., Song, R., Bucknall, R., 2017. Predictive navigation of unmanned surface vehicles in a dynamic maritime environment when using the fast marching method. Int J Adapt Control Signal Process 31, 464–488.

Lyu, H., Yin, Y., 2019. COLREGS-Constrained Real-Time Path Planning for Autonomous Ships Using Modified Artificial Potential Fields. Journal of Navigation 72, 588–608. https://doi.org/10.1017/S0373463318000796

Lyu, H., Yin, Y., 2018. Fast path planning for autonomous ships in restricted waters. Applied Sciences (Switzerland) 8, 6–8. https://doi.org/10.3390/app8122592

Pietrzykowski, Z., Uriasz, J., 2009. The ship domain--a criterion of navigational safety assessment in an open sea area. The Journal of Navigation 62, 93–108.

Rawson, A., Rogers, E., Foster, D., Phillips, D., 2014. Practical application of domain analysis: Port of London case study. The Journal of Navigation 67, 193–209.

Tam, C., Bucknall, R., 2013. Cooperative path planning algorithm for marine surface vessels. Ocean Engineering 57, 25–33. https://doi.org/10.1016/j.oceaneng.2012.09.003

Wang, N., 2010. An intelligent spatial collision risk based on the quaternion ship domain. The Journal of Navigation 63, 733–749.

Zhang, J., Liu, J., Hirdaris, S., Zhang, M., Tian, W., 2023. An interpretable knowledge-based decision support method for ship collision avoidance using AIS data. Reliab Eng Syst Saf 230, 108919. https://doi.org/10.1016/j.ress.2022.108919